

MATHEMATICAL READER

Heat correlations and the full physical complement in SU(2) lattice Yang-Mills theory

Fourth-order coefficients, volume-independent bounds, and original state and energy metrics

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A mathematical introduction followed by all five complete proof manuscripts. The original link coordinates, Haar pairings, physical time and complementary states are retained. The results concern a stated strong-coupling domain and spatial volume at fixed lattice spacing; a four-dimensional continuum mass gap is not claimed.

The earlier workbench's distinct S^6 and Jacobian inputs are credited to Levent Alpöge and their named collaborators in the introduction. The present heat transfer's precise dependencies are identified separately.

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1 Reading the heat and volume results

The basic question of this edition is concrete. A plaquette observable probes a small square of a lattice gauge field. How does its correlation with another plaquette decay under the interacting heat flow? Can one control the error in a fourth-order calculation without that error growing with the size of the lattice? Finally, when states built from those observables are allowed to mix with every other physical state, how much can their energy change?

The five manuscripts collected here answer these questions on an explicitly specified strong-coupling domain. They give the complete fourth-order heat coefficients, a bound on their all-order error that is independent of spatial volume, and quantitative control of the entire complementary physical space. The volume-independent estimates also construct an infinite spatial lattice at **fixed lattice spacing**. They do not establish a four-dimensional continuum Yang-Mills field or its continuum mass gap.

This introduction states the objects and results and explains how the proofs fit together. Appendices A-E reproduce all five supplied proof texts, including their original equation labels, parameter restrictions, and source citations. The exact coefficient tables, proofs, replay scripts and historical records are in the [public workbench](#). This is a readable companion to those records, not a replacement for the complete calculation.

1.1 The original lattice and its physical Hilbert space

Let $L \geq 2$ be an integer. The vertices are $\{-L, \dots, L\}^3$; all contained positive nearest-neighbour links and elementary plaquettes are retained. A link carries a matrix $U_e \in \text{SU}(2)$. The Hilbert space is the subspace of the product-Haar L^2 space invariant under gauge transformations at every vertex, including the boundary vertices. Its inner product is conjugate-linear in the first entry.

For $p = (n; i, j)$, $i < j$, the plaquette observable is the fundamental trace

$$W_p = \text{tr}(U_i(n)U_j(n + e_i)U_i(n + e_j)^{-1}U_j(n)^{-1}).$$

With $T_\alpha = -i\sigma_\alpha/2$ and the corresponding original link derivatives $X_{e,\alpha}$, set

$$K_L = -\sum_{e,\alpha} X_{e,\alpha}^2, \quad H_L = \kappa K_L + \kappa \xi \sum_{p \in \mathcal{P}_L} (2 - W_p), \quad \kappa = \frac{2g^2}{a}, \quad \xi = \frac{1}{4g^4}.$$

Here $a > 0$ is lattice spacing and $g > 0$ is coupling. The operator and form domains are the physical parts of H^2 and H^1 on the finite product of link groups. The smooth positive unit ground vector is ψ_L , its energy is $E_{0,L}$, and

$$A_L = H_L - E_{0,L}, \quad \rho_L = \psi_L^2, \quad r_p = (W_p - \langle W_p \rangle_{\rho_L})\psi_L.$$

Thus r_p is centered in the *interacting* vacuum, not just in Haar measure. On the centered physical space $\mathcal{H}_{0,L} = \psi_L^\perp$, the connected heat correlation is

$$\widehat{C}_{L,pq}(\tau; \xi) = \langle r_p, e^{-\tau A_L / \kappa} r_q \rangle, \quad \tau = \kappa t.$$

The physical time is t ; τ is its stated dimensionless coordinate. The number of plaquettes is $M_L = 3(2L)^2(2L + 1)$, so $M_2 = 240$.

1.2 What the fourth-order calculation contains

The link-center symmetry makes the two-mark correlation even in the homogeneous source ξ . The calculation therefore has the form

$$\widehat{C}_L(\tau; \xi) = C_{0,L}(\tau) + \xi^2 C_{2,L}(\tau) + \xi^4 C_{4,L}(\tau) + \mathcal{R}_{6,L}(\tau; \xi).$$

Appendix A, H2-H4, computes the coefficients from the actual ground-state and resolvent recurrences. The ground-energy shift, division by the original vacuum norm, and subtraction of the ground pole are included. Each connected Laplace coefficient is a finite partial-fraction sum

$$F_\nu(z) = \sum_{c>0, m \geq 1} \frac{a_{c,m}}{(z+c)^m}, \quad C_\nu(\tau) = \sum_{c,m} \frac{a_{c,m}}{(m-1)!} \tau^{m-1} e^{-c\tau}.$$

There are 84 exact time functions across 17 geometric cases. Their **Laplace transforms** are rational functions; the time functions are finite polynomial-exponential sums. The phrase "rational time functions" in the preserved source should be read with that distinction. On the $L = 2$ box the resulting coefficient matrices have respectively 240, 2,496 and 9,660 nonzero entries at degrees zero, two and four. Their time integrals return the earlier static response matrices.

The finite first excitation band is a related, but separate, result. If Y_ξ is the actual projected plaquette frame, its state Gram $G_{\text{band}} = Y_\xi^* Y_\xi$ need not be the identity. Appendix A, H8, gives

$$A_{\text{band}} = 3I + \xi^2 T_2 + \xi^4 T_4 + \dots, \quad T_2 = \frac{7}{15}I - \frac{D_{\text{degree}} + A_{\text{adj}}}{21},$$

$$T_4 - T_4^* = T_2 G_{\text{band},2} - G_{\text{band},2} T_2.$$

This last identity explains why an ordinary coordinate adjoint is not the adjoint in the computed band metric. On the open $L = 2$ box its right-hand side has 1,440 nonzero entries. The band remainder is proved only for $|\xi| < 1/(32M_L)$; the later volume-independent *heat* result does not silently enlarge that band domain.

1.3 From a finite-box estimate to a volume-independent heat bound

Appendix B first proves an entrywise analytic bound on $|\xi| < 1/M_L$. This controls a fixed box, but its source radius shrinks as the box grows. Appendix D addresses exactly that dependence. The proof uses a locally labelled Fourier-coefficient norm to construct the exponential source, while keeping the original Haar pairing and all scalar energy terms.

The essential estimates are as follows. The physical representation labels satisfy $c_j \geq 6j_e$ at each active link and $c_j \geq 3$ away from the constant block. The original plaquette coefficient has trace norm 8. These facts give a convergent labelled source series on

$$R_0 = \frac{3}{256}, \quad \chi(r) = \frac{1 - \sqrt{1 - 256r/3}}{2}, \quad d(r) = 3(1 - \chi(r)).$$

The scalar mean is solved together with the nonconstant equation. A closed generator on a stated dense coefficient domain then gives the actual heat flow, rather than replacing a semigroup estimate by an estimate on one inverse. The Poisson identity pairs that flow against an arbitrary signed sum of plaquettes. Taking the supremum over the signs controls an entire matrix row.

For $\|B\|_{\text{row}} = \max_p \sum_q |B_{pq}|$, the result is

$$\|\mathcal{R}_{6,L}(\tau; \xi)\|_{\text{row}} \leq 512e^{-3\tau/2} \frac{\theta^6}{1 - \theta^2}, \quad \theta = \frac{|\xi|}{R_0} < 1.$$

It holds for every $L \geq 2$ and $\tau \geq 0$ with the same constants (Appendix D, U29-U30). A second bound, U29b, has the smaller radius $R_1 = 45/4096$, prefactor $6184/25$, and decay $15/8$. Both bounds are retained with their own radii; the better of their scalar error bounds may be used where both apply.

Local Taylor coefficients agree once their original support fits inside both boxes. Combined with the complete analytic tail, this gives full-sequence convergence of local vacuum means and heat correlations. U9 constructs the limiting gauge-invariant state and symmetric Markov semigroup, with a self-adjoint generator on its physical L^2 space. This passage increases spatial volume with a fixed. It is not an ultraviolet limit $a \rightarrow 0$.

1.4 The three Grams and why the complement matters

Let $R : \mathbb{C}^{M_L} \rightarrow \mathcal{H}_{0,L}$ be the column map $Rc = \sum_p c_p r_p$, and define its inverse-energy family

$$\Phi = \kappa A_L^{-1} R, \quad A_L \Phi = \kappa R.$$

The three original Grams measure different things:

$$G_0 = R^* R, \quad G_1 = R^* \Phi, \quad G_2 = \Phi^* \Phi, \quad E = \Phi^* A_L \Phi = \kappa G_1.$$

Here G_0 is the norm of the forcing columns, G_2 is the norm of the inverse-energy states, and E is their energy form. More generally, $G_k = \kappa^k R^* A_L^{-k} R$ for $k \geq 1$. The heat integral computes all these inverse-energy moments:

$$G_k = \frac{1}{(k-1)!} \int_0^\infty \tau^{k-1} \widehat{C}_L(\tau; \xi) d\tau \quad (k \geq 1).$$

At $g^2 \geq 16$, Appendix E, M10, obtains, uniformly over $L \geq 2$ and $a > 0$,

$$\|G_0 - I\| < \frac{124}{10^6}, \quad \|G_1 - I/3\| < \frac{66}{10^6}, \quad \|G_2 - I/9\| < \frac{36}{10^6}.$$

The coefficient bound is valid at boundary rows because absolute values are summed *before* contributions from distinct supports can cancel. The full-space estimate is $A_L \geq (117/40)\kappa I$ on $\mathcal{H}_{0,L}$.

These facts alone would not justify pretending that $\text{im } \Phi$ is a reducing subspace. Its coupling to the rest of the Hilbert space must be retained. Define

$$P = \Phi G_2^{-1} \Phi^*, \quad Q = I - P, \quad B = QR.$$

Then P is the actual orthogonal state projection and

$$B^* B = G_0 - G_1 G_2^{-1} G_1.$$

For every form-domain vector written as $\Phi c + h$, $h \in Q\mathcal{H}_{0,L}$, the full energy is

$$q_{A_L}(\Phi c + h) = c^* E c + 2\kappa \text{Re}\langle Bc, h \rangle + q_{A_L}(h).$$

Appendix E, M5, proves that the complementary operator $D_Q = QA_LQ$ on $\text{Dom}(A_L) \cap Q\mathcal{H}_{0,L}$ is self-adjoint and bounded below. Minimization over its entire form domain therefore gives

$$h_{\min}(c) = -\kappa D_Q^{-1}Bc, \quad E_{\text{full}} = E - \kappa^2 B^* D_Q^{-1}B,$$

$$G_{\text{restored}} = G_2 + \kappa^2 B^* D_Q^{-2}B.$$

At $g^2 \geq 16$ the sharpened bounds are

$$\frac{999}{1000}E \prec E_{\text{full}} \preceq E, \quad G_2 \preceq G_{\text{restored}} \prec \frac{1001}{1000}G_2.$$

Thus the original family's energy decreases by less than one part in a thousand when the entire complement is allowed to relax, while its state Gram increases by less than one part in a thousand. The inequalities are between quadratic forms in the same original coefficients; a strict inequality is evaluated on nonzero coefficients. They are not quotients of matrix entries. M8 carries the bounded maps and estimates to ℓ^2 of the infinite plaquette family, including its full physical complement.

1.5 Two concrete consequences

First, take opposite faces $p = (0, 0, 0; 0, 1)$ and $q = (0, 0, 1; 0, 1)$. Their degree-zero and degree-two correlations vanish; the degree-four term consists of four original three-face paths and one cube. At $\xi = 10^{-10}$, so that $g^2 = 50000$ and $\kappa = 100000/a$, Appendix E, M26, gives

$$\frac{85910}{10^7} < \frac{C_{pq}(1/\kappa; \xi)}{\xi^4} < \frac{85920}{10^7}.$$

The complete correlation is positive throughout $1/\kappa \leq t \leq 3/\kappa$, in every $L \geq 2$ and in the fixed-spacing spatial limit. The all-order remainder is smaller than the explicitly proved coefficient lower bound on this interval. Positivity is therefore a statement about the actual interacting correlation, not merely its truncated expansion.

Second, finite heat time gives a controlled replacement of inverse-energy states:

$$\Phi_T = (I - e^{-TA_L})\Phi = \kappa \int_0^T e^{-tA_L} R dt, \quad A_L \Phi_T = \kappa R - \kappa e^{-TA_L} R.$$

The omitted forcing is explicit. If $\varepsilon = e^{-\kappa dT}$ with the full-space lower bound $A_L \geq \kappa dI$, spectral calculus gives

$$(1 - \varepsilon)^2 G_2 \preceq \Phi_T^* \Phi_T \preceq G_2, \quad (1 - \varepsilon)^2 E \preceq \Phi_T^* A_L \Phi_T \preceq E.$$

These inequalities survive restriction and minimization over the *same coefficient fibers*. In particular, a signed sum $\mathcal{L} = \sum_j a_j \log \det Q_j$ of such original positive quotient forms satisfies

$$|\mathcal{L}_T - \mathcal{L}| \leq 2\mathcal{B}[-\log(1 - \varepsilon)], \quad \mathcal{B} = \sum_j |a_j| \text{rank } Q_j.$$

For four returns with coefficients $a_j \in \{-1, +1\}$ and ranks at most 240, so that $\mathcal{B} \leq 960$, $g^2 \geq 16$ and $T = 10/\kappa$ give error below 4×10^{-10} in every exterior volume. For a growing observation family, M25 keeps the rank explicitly and reaches tolerance $\eta > 0$ at

$$T = \frac{1}{\kappa d} \log \left(1 + \frac{2\mathcal{B}}{\eta} \right).$$

1.6 Sources, mathematical lineage, and how to verify the edition

The exponential-vacuum and character framework has a human antecedent in Schütte, Zheng Weihong and Hamer [1]. The block minimization is a Feshbach-Schur construction; [2] is a modern primary reference. Appendix D proves the needed compact-group Fourier coefficient estimates and cites Eymard [3]. Its generator argument states the domain, dissipativity and range conditions used from Lumer and Phillips [4]. These are point-of-use dependencies, not generic acknowledgments.

Two distinct contributions associated with **Levent Alpöge** enter the earlier workbench and deserve separate attribution. The S^6 branch starts from the claimed complex-threefold construction he circulated with AI assistance [5], also credited to him in Engel's exposition [6]. A different branch starts from his Jacobian counterexample: global noninjectivity despite a constant nonzero Jacobian. His original announcement [7] credits Akhil's question and Fable's work; Tao's later exposition is credited separately [8]. These constructions are not interchangeable. Their subsequent coordinate maps and conductor calculations belong to their respective continuations. The five heat manuscripts collected here do not directly use either originating construction as a lemma: their actual transferred inputs are the heat, Gram, quotient-minimum and inverse-power maps specified next. This distinction records the programme's mathematical origins without attributing the present heat bounds to Alpöge or claiming an unproved spectral equivalence.

The workbench's [attribution](#) and [mathematical lineage record](#) gives the primary sources and exact historical points of use.

The direct transfer used in the present proofs is the heat/Gram and minimum-fiber argument from the Riemann workbench, not a spectral identification of its arithmetic objects with Yang-Mills fields. Appendix C identifies the receiving maps explicitly. Its pinned source revision is `128aa308dd2a70ba6e073816a957d1ee6414ba2c` in the [Riemann workbench](#): HM6-HM14 and HM19-HM24 of `HEAT_TO_ORIGINAL_QUOTIENT_METRICS.tex`, HG6-HG10 of `ORIGINAL_HEAT_AND_SOURCE_ENDPOINT.tex`, and IK9-IK12 of `INVERSE_POWER_CONDUCTOR_OBSERVABILITY.tex`. The source intake records retain their full paths and blob identities. Original conductor coefficients, zeta-zero data and arithmetic constants remain attached to their source constructions.

The workbench was developed with AI assistance from ChatGPT 5.6 Sol in Ultra mode in Codex and GPT-6 Astra; the collection preserves the individual mathematical and source records. The supplied mathematical arguments are presented for scrutiny. Fresh integration checks repeated 2,425 exact volume checks with 16 false-formula controls, 879 heat checks with 21 controls, and 52 route calculations, in both ordinary and optimized Python. The cumulative manifest passed before and after these runs. These executions verify their declared finite algebra and endpoints. They are not Lean proofs, independent external peer review, or a formal verification of every infinite-dimensional analytical step.

The source texts remain complete below, so the numerical statements can be followed through their constructions rather than accepted from this introduction. The original cumulative volume archive contains 2,015 files and retains all files of the preceding heat edition, including the five earlier control documents in its history directory. A non-mutating integrity check from the extracted archive root is `python -B verify_cumulative.py`. Full producer replays should use a new disposable copy; their scripts generate files and are not read-only checks.

The mathematical advance described here is the passage from finite-box heat coefficients to a volume-independent heat error and to a quantitative full-complement return in original physical metrics. The next, different issue is extension beyond the stated strong-coupling domain. The simultaneous shrinking-spacing path discussed in the source eventually leaves that domain. Neither the existing local source certificate nor its elementary restart establishes the ultraviolet continuum limit.

1.7 References

1. D. Schütte, Zheng Weihong and C. J. Hamer, *The Coupled Cluster Method in Hamiltonian Lattice Field Theory*, 1996, [arXiv:hep-lat/9603026](#).
2. G. Dusson, I. M. Sigal and B. Stamm, *The Feshbach-Schur map and perturbation theory*, 2021, [arXiv:2105.02058](#).
3. P. Eymard, *L'algèbre de Fourier d'un groupe localement compact*, Bulletin de la Société Mathématique de France 92 (1964), 181-236. [Original article](#).
4. G. Lumer and R. S. Phillips, *Dissipative operators in a Banach space*, Pacific Journal of Mathematics 11 (1961), 679-698, Theorem 3.1, pp. 686-687. [Original article](#).
5. Levent Alpöge, S^6 construction circulated with AI assistance, [original 108-page manuscript](#), especially p. 2, Setup, for the coordinate matrix Π .
6. Philip Engel, *Complex structures on S^6* , 13 September 2026, [expository manuscript](#), abstract and Section 1.3.
7. Levent Alpöge, [original announcement of the Jacobian counterexample](#), 20 July 2026. The announcement credits Akhil and Fable.
8. Terence Tao, *A digestion of the Jacobian conjecture counterexample*, 21 July 2026, [exposition](#).

A Original Yang–Mills heat correlations: the complete fourth-degree return

21 September 2026. This is an additive continuation of the recovered cumulative workbench. The original open-box SU(2) operator, physical units, Haar measure, gauge constraints and source coordinates remain. The coefficient calculation uses the complete available fifth-source archive. The separately saved sixth-source proofs are preserved with their recovery status; their missing catalogues are not prerequisites for this calculation. General exponential-vacuum and character methods retain Schütte–Zheng–Hamer (1996) as a human antecedent. The exact finite calculations here are accompanied by analytic proofs and executable audits; independent external analytical review and historical priority are not asserted.

A.1 H1. The actual operator and time coordinate

For $L \geq 2$ use every vertex n in $\{-L, \dots, L\}^3$, every contained positive edge (n, i) , and every contained elementary face $p = (n; i, j)$, $i < j$. Put

$$\begin{aligned} W_p &= \text{tr}(U_i(n) U_j(n+e_i) U_i(n+e_j)^{-1} U_j(n)^{-1}), \\ T_\alpha &= -i \sigma_\alpha / 2, \\ X_{e, \alpha} &= \frac{d}{ds} f(\dots, \exp(s T_\alpha) U_e, \dots) \text{ at } s=0, \\ K &= -\sum_e (X_{e, \alpha})^2, \\ H(x) &= \kappa K + \kappa \sum_p x_p (2 - W_p), \\ \kappa &= 2g^2/a, \quad x_p = xi/(4g^4) \text{ on the homogeneous line.} \end{aligned} \quad (H1)$$

The inner product on the original product Haar probability space is conjugate linear in its first entry. The physical subspace is invariant under all vertex gauge transformations, including boundary vertices. The domains are the physical parts of H^2 and H^1 . The original bounded smooth potential preserves these domains; the compact elliptic operator has a smooth strictly positive unit ground state ψ_x with energy $E_0(x)$. These domain and vacuum facts are the ones in the pinned original finite-box source (SOURCE_INTAKE.json, YM0).

The coefficient operator and its inverse physical return are exactly

$$\begin{aligned} B_x &= (H(x) - 2\kappa \sum_p x_p I) / \kappa = K - \sum_p x_p W_p, \\ H(x) &= \kappa B_x + 2\kappa \sum_p x_p I, \\ E_0(x) &= \kappa [2 \sum_p x_p + e(x)]. \end{aligned} \quad (H2)$$

Both scalar energy terms remain in the return. Let $A_x = H(x) - E_0(x)$ and $r_p = (W_p - \langle W_p \rangle_{\psi_x}) / \rho$, $\rho = \psi_x^2$. Define

$$\begin{aligned} C_{pq}(t; x) &= \langle r_p, \exp(-t A_x) r_q \rangle, \\ \text{Chat}_{pq}(\tau; x) &= C_{pq}(\tau / \kappa; x), \\ \tau &= \kappa t, \quad t = \tau / \kappa. \end{aligned} \quad (H3)$$

This is an invertible physical-time change; no link or field coordinate is changed. The Laplace variable z below is conjugate to this stated τ :

$$\begin{aligned} F_{pq}(z; x) &= \int_0^\infty \exp(-z \tau) \text{Chat}_{pq}(\tau; x) d\tau \\ &= \kappa \langle r_p, (A_x + \kappa z)^{-1} r_q \rangle. \end{aligned} \quad (H4)$$

All expressions are initially at real sufficiently small x and $z > 0$. The separate analytic proof gives their full domains and uniform time remainders.

A.2 H2. The complete coefficient recurrence, including the ground pole

Retain the source section $u = \psi_x / m_x$, $m_x = P_H \psi_x$. Its inverse is $\psi_x = m_x u$, and its original scalar obeys $m_x^2 \langle u, u \rangle_H = 1$. Thus $P_H u = 1$ and the original norm $N(x) = \langle u, u \rangle_H$ is kept in every expectation. Expand $u(x) = \sum_{\nu} u_{\nu} x^{\nu}$, $e(x) = \sum_{\nu} m_{\nu} (u_{\nu} = 0) e_{\nu} x^{\nu}$, with $u_0 = 1$, $P_H u_{\nu} = 0$ for $\nu \neq 0$. The exact eigen-equation gives

$$\begin{aligned} e_{\nu} = & -\sum_{(j: \nu_j > 0)} P_H(W_{-j} u_{(\nu - e_j)}), \\ K u_{\nu} = & \sum_j W_j u_{(\nu - e_j)} \\ & + \sum_{(0 < \mu \leq \nu)} e_{\mu} u_{(\nu - \mu)}. \end{aligned} \quad (H5)$$

The last term includes $e_{\nu} u_0$, and its Haar scalar makes the RHS centered. Its unique centered inverse is the original $K^{-1} Q_H$. The recovered coefficient engine supplies these actual finite coefficients and all their source labels.

For a marked face q , define $h_q(z; x) = (B_x - e(x) + z)^{-1} W_q u(x)$. Its coefficient equation is

$$\begin{aligned} (K+z) h_{(q, \nu)} = & W_q u_{\nu} + \sum_j W_j h_{(q, \nu - e_j)} \\ & + \sum_{(0 < \mu \leq \nu)} e_{\mu} h_{(q, \nu - \mu)}. \end{aligned} \quad (H6)$$

The source index may include repeated faces; all multiplicities remain literal multiindex coefficients. Define

$$\begin{aligned} N_{\nu} = & \sum_{(\alpha + \beta = \nu)} \langle u_{\alpha}, u_{\beta} \rangle_H, \\ M_p_{\nu} = & \sum_{(\alpha + \beta = \nu)} \langle u_{\alpha}, W_p u_{\beta} \rangle_H, \\ H_{pq}_{\nu} = & \sum_{(\alpha + \beta = \nu)} \langle u_{\alpha}, W_p h_{(q, \beta)} \rangle_H. \end{aligned} \quad (H7)$$

Coefficient division by N , with $N_0 = 1$, gives the original mean $\mu_p = M_p / N$ and the uncentered resolvent H_{pq} / N . The complete connected answer is

$$F_{pq} = H_{pq} / N - \mu_p \mu_q / z. \quad (H8)$$

The subtracted term is retained as an actual ground-state contribution. The source shift e_{μ} in (H6), both norm divisions and this term are all required. Every resulting coefficient in the catalogue has zero coefficient at each power z^{-j} , $j > 0$. The original positive free excitation gap also proves this cancellation analytically near $(x, z) = (0, 0)$, after the ground projection is removed.

A.3 H3. Exact original Casimir inversion and time reconstruction

An original invariant trace monomial has a finite tensor representation on its original edge set. For r_e occurrences on edge e , the allowed doubled spins are $r_e, r_e - 2, \dots$. At each original vertex the sum of the doubled spins is even, and no one exceeds the sum of the others. These are the SU(2) invariant-tensor conditions: iterated angular-momentum addition supplies every intermediate spin in the parity-compatible interval, so zero belongs exactly on those conditions. The resulting finite Casimir list contains all eigenvalues of that monomial. It retains multiplicities inside every equal-eigenvalue eigenspace.

For its distinct values C , use the literal original operator projectors

$$\begin{aligned} P_c = & \text{product}_{(d \in C, d \neq c)} (K - dI) / (c - d), \\ (K+z)^{-1} f = & \sum_{(c \in C)} P_c f / (z + c). \end{aligned} \quad (H9)$$

The full physical operator has all spins. The finite list in (H9) is the complete representation space reached by that Taylor coefficient and its marked word. It does not discard a path contributing to the coefficient.

Every response coefficient is stored as exact partial fractions

$$F_{\nu}(z) = \sum_{(c>0, n \geq 1)} a_{\nu}(c, n) / (z+c)^n. \quad (H10)$$

Their time functions are exactly

$$\text{Chat}_{\nu}(\tau) = \sum_{(c, n)} a_{\nu}(c, n) \tau^{n-1} \exp(-c \tau) / (n-1)!. \quad (H11)$$

Indeed direct integration of each finite term gives (H10). The analytic heat construction in the companion proof shows these are the actual Taylor coefficients, and uniqueness of the Laplace transform identifies them. For explicit uniqueness, weighting a vanishing transform by $\exp(-s_0 \tau)$ produces a finite complex measure; $x = \exp(-\tau)$ carries its integer Laplace samples to all polynomial moments on $[0, 1]$. Polynomial approximation forces the measure to vanish; continuity gives equality of the time functions.

Products in the partial-fraction engine are also exact. For $a \neq b$, the principal part of $(z+a)^{-m}(z+b)^{-n}$ at $-a$ has coefficients

$$(-1)^k \binom{n+k-1}{k} / (b-a)^{n+k}, \quad k=0, \dots, m-1, \quad (H12)$$

at power $(z+a)^{-(m-k)}$; the analogous $-b$ principal part is retained. Their sum has the same two principal parts and vanishes at infinity, so the rational functions are equal. Coincident poles add their orders. This retains every repeated pole and hence every time-polynomial factor in (H11).

The exact original source equation (H6) is audited as a complete polynomial in all retained tree/chord quaternion coordinates, separately for every pole. The map is $Z_e = t_s U_e t_t^{-1}$, with its original inverse $U_e = t_s^{-1} Z_e t_t$ and the tree variables retained. On each chord, $Z = x_0 \prod_i x_a \sigma_a$ and $\sum_{(a=0)}^3 x_a^2 = 1$. The remainder basis has x_0 exponent zero or one. The map and injectivity argument are the recovered fifth-source coordinate proof, unchanged. Every residual coefficient is zero in this full product-of-spheres algebra, not merely at sampled links.

A.4 H4. Complete support classification and original box assembly

A change $U_e \rightarrow -U_e$ multiplies each source or marked trace using e by -1 . Thus a nonzero coefficient has an even total multiplicity on every original edge, counting both marked traces. At total face multiplicity 2 or 4 this requires all face multiplicities even. At total multiplicity 6 it permits, in addition, the six distinct faces of one elementary cube. The original closed cubical two-chain argument is retained in the sixth-energy proof: a nonempty closed mod-two face set has at least six faces; at six, each of the four edges of an extreme face requires its adjacent side face, and closure of their remaining edges forces the opposite face. The resulting set is the boundary of that original cube.

A disconnected face source factors over disjoint original edge factors. The original operators, positive vacua and their means factor accordingly. A source component containing neither mark cancels between numerator and N ; marks in different components have zero connected correlation. This remains true with shared graph vertices: the full tensor-product Hamiltonian

and the actual invariant vectors give the same physical expectation. Connectedness therefore means sharing original edges, exactly as in the source catalogue.

The complete list is:

heat/source degree	total face multiplicity including marks	coordinate cases	marked responses
0	2	1	1
2	4	3	7
4	6	13	76

There are 84 full rational time functions. At total degree six the cases are one single face, two geometries of an ordered 4+2 pair, seven geometries of a three-face path, one common-edge triple, one cube corner and one cube. All marks and all allowed diagonal marks are retained. Faces with repeated multiplicity are never replaced by a factorial convention.

A signed coordinate permutation and translation

$$\begin{aligned} y_i &= \epsilon_{pi(i)} x_{(pi(i))} - d_i, \\ x_{(pi(i))} &= \epsilon_{pi(i)} (y_i + d_i) \end{aligned} \quad (H13)$$

transports every actual face multiset and both marked faces to its record. Each original edge is mapped to the corresponding forward or inverse edge. The product-Haar map is unitary, carries K to K on that support, and maps the actual face words to the target face word or its inverse. In $SU(2)$, $\text{tr}(U^{-1}) = \text{tr}(U)$ by the two eigenvalues λ, λ^{-1} ; hence the same trace is returned. The inverse in (H13) returns every source to its original coordinates.

The $L=2$ assembly retains all 240 original faces, 1,128 adjacent pairs, 6,732 paths, 576 common-edge triples, 512 corners and 64 cubes. The coefficient matrices have respectively 240, 2,496 and 9,660 nonzero entries. Their entire time integrals agree entry by entry with the preceding static matrices:

$$\begin{aligned} \text{int}_{0^{\infty}} \text{Chat}_0(\tau) d\tau &= I/3, \\ \text{int}_{0^{\infty}} \text{Chat}_2(\tau) d\tau &= -(5/36)I + (2/1053)D_{\text{degree}} + (4/1053)A_{\text{adj}}, \end{aligned} \quad (H14)$$

and the full original Chat_4 integral is the previously calculated R_4 , including every boundary row and cube. All complete matrices and all their coordinate transports are in `generated/heat_matrix_L2.json` and `heat_transports_L2.json`. Those files preserve the original face ordering.

A.5 H5. Explicit time-dependent functions

For one original plaquette with only its own source parameter, the degree-two coefficient is

$$\exp(-3\tau) [-241/900 - (7/15)\tau] + (4/225)\exp(-8\tau). \quad (H15)$$

For two different faces sharing one original edge, the coefficient of $x_p x_q$ is

$$\begin{aligned} \exp(-3\tau) [-409/17199 + \tau/21] \\ + \exp(-9\tau/2)/36 + \exp(-13\tau/2)/588. \end{aligned} \quad (H16)$$

Its integral is $4/1053$ and its value at zero is $2/351$. The corresponding diagonal, neighbour-source term replaces $-409/17199$ by $-13/441$; its integral is $2/1053$ and its time-zero value is zero. These distinct values retain the full source and state pairings.

Now take opposite faces

$$p=(0,0,0;0,1), \quad q=(0,0,1;0,1). \quad (\text{H17})$$

Their coefficients at degrees zero and two vanish. Four original three-face paths and one original cube contribute at degree four. One path gives

$$\begin{aligned} P(\tau) = & \exp(-3\tau) [75173779/67612708800 \\ & - (4441/3611790)\tau + \tau^2/882] \\ & - (121/108864)\exp(-9\tau/2) + (1/11664)\exp(-6\tau) \\ & - (197/9988160)\exp(-13\tau/2) + (1/163800)\exp(-8\tau) \\ & + (1/6492304)\exp(-10\tau). \end{aligned} \quad (\text{H18})$$

The cube gives

$$\begin{aligned} B(\tau) = & \exp(-3\tau) [-235/648 + (10/27)\tau] \\ & + \exp(-9\tau/2) [1123/2916 + (23/81)\tau + \tau^2/18] \\ & + (1/648)\exp(-6\tau). \end{aligned} \quad (\text{H19})$$

Thus the complete actual degree-four coefficient is

$$\begin{aligned} h_4(\tau) &= 4P(\tau) + B(\tau), \\ h_4(0) &= 8869/365040, \\ \int_0^\infty h_4(\tau) d\tau &= 641033/29568240. \end{aligned} \quad (\text{H20})$$

All six decay rates and both repeated-pole orders remain. The coefficient at $\tau=1$ is enclosed by rational exponential bounds in generated/heat_bounds.json; its numerical size is approximately 0.00859150755. This diagnostic decimal is not an acceptance test. The companion proof turns (H20) into intervals for the full actual correlation, retaining every higher Taylor order.

A.6 H6. Independent checks on the same original geometry

The single-plaquette check uses the infinite original character basis $\chi_j(j/2)$, where $K \chi_j(j/2) = j(j+2)\chi_j(j/2)$ and $W \chi_j(j/2) = \chi_{j-1}((j-1)/2) + \chi_{j+1}((j+1)/2)$, with the negative-index term zero. At order n every contributing insertion path has $j \leq n+1$; the independent recurrence retains all these indices. It reproduces all three self-source heat coefficients, with both original norm divisions and the ground pole. Its integrated coefficients are $1/3, -5/36, 289/5184$. They also follow by twice differentiating the retained one-plaquette energy coefficients. The original radial coordinate return $e_{\text{one}}(\xi) = b_2(-4\xi)/4-1$ matches the NIST DLMP 28.6.5 expansion; the map and scalar energy term are in the recovered audit report. This comparison covers that one-plaquette coefficient convention only.

There is an additional computation for every marked cube pair. Orient the six faces outward. Original Haar integration yields $2^8/2^{12} = 1/16$. For any proper visited face subset S , each edge already visited twice can never occur in a later insertion. Its surviving representation is the original spin zero; each boundary edge carries the remaining spin one-half. Therefore its original intermediate energy is $c(S) = 3|S|/4$.

For marks $p!=q$, arrange the other four faces in a permutation and split the ordered list into right-vacuum, middle-heat and left-vacuum pieces. All 24 permutations and all 15 length triples occur.

Right and left vacuum prefixes contribute $1/c(S)$; the middle heat state begins at $\{q\}$ union right and each middle insertion contributes $1/[z+c(S)]$. The original $1/16$ remains. Summing these 360 rational functions gives exactly the cube coefficient in (H10). All fifteen marked cube pairs were checked: 5,400 complete original insertion records, including every intermediate boundary energy. This calculation imports none of the trace-differentiation or Casimir-projection engine.

The time-zero covariance, its first derivative $-\langle \text{Gamma}(W_p, W_q) \rangle_{\rho}$, and its second derivative $\langle [K, W_p]u, [K, W_q]u \rangle_{H/N}$ are also calculated separately from the entire original density quotient at each coefficient. All 84 rational functions reproduce these three values. The full-polynomial source audit and these moment identities have distinct execution records.

A.7 H7. Sources and present mathematical scope

The original operator/domain and maximal-tree maps are the pinned finite-box workbench source. The exponential and character/Casimir antecedent is D. Schütte, Zheng Weihong and C. J. Hamer, arXiv:hep-lat/9603026. The residue and minimum-section comparison imported from the current Split-Zero heat work is specified completely in RH_HEAT_TRANSFER.md, with its exact source revision and read equations. No arithmetic period or Gamma constant is assigned to a physical Yang-Mills matrix.

These are actual finite-regulator heat coefficients and actual finite-volume analytic bounds. The Cauchy radius in the companion is $1/M$. It therefore retains dependence on the original volume, and does not supply an ultraviolet continuum construction. The earlier uniform-gap manuscripts retain their recorded independent-audit qualification.

A.8 H8. The complete fourth-order first-band return in its original metric

The original free energy-three space is exactly $\text{span}\{W_p\}$. To see the next separation, an active invariant graph with four edges is a four-cycle; its vertex intertwiners force equal edge spins, giving energy $4j(j+1)=3,8,\dots$. A simple bipartite active graph with five edges and minimum degree at least two would have one cyclic component with five vertices and five edges, hence an odd five-cycle, or at most four vertices, where bipartiteness permits at most four edges. Both cases contradict its assumptions. Therefore any other active assignment has at least six edges and energy $\geq 9/2$. Thus

$$\begin{aligned} \text{spec}(K_{\text{phys}}) &\subset \{0,3\} \cup [9/2, \text{infinity}), \\ P3H_{\text{phys}} &= \text{span}\{W_p\}, \quad \dim P3 = M. \end{aligned} \tag{H21}$$

For sufficiently small real ξ let P_{ξ} be the complete spectral projection near energy 3 of $B_{\xi-e}(\xi)$, and put

$$\begin{aligned} Y_{\xi} &= P_{\xi} \sum_p c_p r_p, \\ G_{\text{band}} &= Y_{\xi} * Y_{\xi}, \\ (B_{\xi-e}) Y_{\xi} &= Y_{\xi} A_{\text{band}}. \end{aligned} \tag{H22}$$

The exact quantitative construction below proves invertibility of this frame onto the actual band. At $\xi=0$, $c \rightarrow \sum c_p W_p$ is the original Haar isometry R_0 . Projection of the heat resolvent onto its original energy-three poles gives

$$\begin{aligned} C_{\text{band}}(\tau) &= G_{\text{band}} \exp(-\tau A_{\text{band}}), \\ G_{\text{band}} &= I + \xi^2 G_2 + \xi^4 G_4 + \dots, \\ A_{\text{band}} &= 3I + \xi^2 T_2 + \xi^4 T_4 + \dots \end{aligned} \tag{H23}$$

The original center symmetry makes both matrices even. For each of the full heat matrices let $A_{(n,j)}$ be the coefficient of $(z+3)^{-j}$. Then literal multiplication of the two series in (H23) proves

$$\begin{aligned} G_2 &= A_{(2,1)}, \quad G_4 = A_{(4,1)}, \quad T_2 = -A_{(2,2)}, \\ A_{(4,3)} &= T_2^2, \\ T_4 &= -A_{(4,2)} - G_2 T_2. \end{aligned} \quad (\text{H24})$$

All matrix factors retain their order. The complete $L=2$ calculation gives

$$T_2 = (7/15)I - (D_{\text{degree}} + A_{\text{adj}})/21. \quad (\text{H25})$$

It gives all 9,660 nonzero entries of T_4 and all 9,660 entries of G_4 in first_band_L2.json. Every entry of $A_{(4,3)} = T_2^2$ is checked. Original self-adjointness returns through the full Gram as

$$\begin{aligned} G_{\text{band}} A_{\text{band}} &= A_{\text{band}} * G_{\text{band}}, \\ T_4 - T_4^* &= T_2 G_2 - G_2 T_2. \end{aligned} \quad (\text{H26})$$

The right side has 1,440 nonzero entries for this open box. For the original faces $(-2,-2,-2;0,1)$ and $(-2,-1,-2;0,1)$, the displayed difference is exactly $2/7371$. This is the explicit correction between the coordinate adjoint and the original metric adjoint. Its value is retained, rather than forcing a symmetric coefficient matrix by dropping the mixed product.

Here is a complete analytic remainder for (H23). Work on the complex homogeneous circle $|\zeta| = 1/(32M)$. The original ground/complement calculation in A1--A2 applies with $\|W\|^2 = \eta = 1/(1024M)$ and $\text{Re } D \geq 47/16$. On $|e| = \eta$, $\|Z(D-e)^{-1}W\| < \eta$, so the same scalar argument principle gives $|e| \leq \eta$. Consequently

$$\|B_{\zeta} - e - K\| \leq 1/16 + 1/(1024M) < 1/15. \quad (\text{H27})$$

On the spectral circle $|\lambda - 3| = 1/2$ the original free resolvent has norm at most 2. Its norm-convergent resolvent series gives

$$\begin{aligned} \|P_i - P_3\| &\leq 2/13, \\ \|(B_{\zeta} - e - 3)P_i\| &\leq 1/13. \end{aligned} \quad (\text{H28})$$

For the first inequality, integrate the difference of resolvents: the bound is $2b/(1-2b)$ with $b \leq 1/15$. For the second, integrate that difference multiplied by $\lambda - 3$, obtaining $b/(1-2b)$. The corresponding free integral is zero because $(K-3)P_3 = 0$. The circle encloses exactly the original M -dimensional band, by the same resolvent homotopy and finite rank at zero.

Use the exact Haar section $u = 1 + w$, where $\|w\| \leq 1/(92\sqrt{M})$; the denominator in its block inverse exceeds $23/8$. The column map $Fu: c \rightarrow \sum c_p W_p u$ satisfies

$$\|Fu - R_0\| \leq 2\sqrt{M} \|w\| \leq 1/46.$$

Define $Y_{\text{hat}} = P_i F_u$. Its original coefficient map obeys

$$\begin{aligned} \|R_0 * Y_{\text{hat}} - I\| &\leq 107/598, \\ \|(R_0 * Y_{\text{hat}})^{-1}\| &\leq 598/491. \end{aligned} \quad (\text{H29})$$

Indeed the two contributions are Pi-P3 and Pi(Fu-R0) , bounded by $2/13$ and $(15/13)/46$. Its inverse on the band is $(\text{R0}^*\text{Yhat})^{(-1)}\text{R0}^*$. The actual unit-vacuum frame is $\text{Y}=\text{m Yhat}$, with $\text{psi}=\text{m u}$ and $\text{m}^2<\text{u,u}>_{\text{H}}=1$ retained. Its scalar cancels from the operator intertwining, not from Gband . Thus

$$\begin{aligned} \text{Aband} &= (\text{R0}^*\text{Yhat})^{(-1)}\text{R0}^*(\text{B}_{\text{zeta-e}})\text{Yhat}, \\ ||\text{Aband}-3\text{I}|| &\leq (598/491)(1/13)(47/46)=47/491. \end{aligned} \quad (\text{H30})$$

All these maps are analytic on a neighborhood of the closed source circle. The original center involution sends $\text{Yhat}_{-\text{zeta}}=-\text{C Yhat}_{\text{zeta}}$ and $\text{R0}^*\text{C}=-\text{R0}^*$, so Aband is even. Cauchy's formula and the full even tail prove

$$\begin{aligned} ||\text{Aband}-3\text{I}-\text{xi}^2\text{T2}-\text{xi}^4\text{T4}|| \\ \leq (47/491)(32\text{M}|\text{xi}|)^6/[1-(32\text{M}|\text{xi}|)^2], \\ |\text{xi}| < 1/(32\text{M}). \end{aligned} \quad (\text{H31})$$

Multiplication by kappa returns this to physical energy. This is an actual finite-volume matrix remainder, with all original band coordinates, the nonidentity Gram (H23), and the dependence on M retained. No diagonalization of T4 or uniform-in-volume fourth-order spectral claim is inferred here.

B Full-time heat bounds and the original plaquette response metrics

21 September 2026. This proof uses the unchanged finite-box SU(2) Hamiltonian and the complete coefficients in PHYSICAL_HEAT_COEFFICIENTS.md. Its analytic argument is finite-volume and independent of the inherited volume-uniform source-norm estimates. The original number M of plaquettes appears in every Cauchy radius. Written arguments and executed finite checks have separate records. All inner products are conjugate-linear in the first entry.

B.1 A1. Original free spectrum and Haar blocks

Use H1--H3 of the coefficient proof. The original physical free operator K has its constant eigenvector 1, and every other physical Fourier block has energy at least 3. Here is the graph argument. In a nonzero invariant representation assignment, a vertex incident to only one active edge has no invariant tensor. Thus the finite active graph has minimum degree at least two and contains a cycle. The original simple cubic graph has no triangles and every cycle has at least four edges. Each active edge has spin $j \geq 1/2$ and Casimir $j(j+1) \geq 3/4$. Hence the complete physical K on the orthogonal complement of 1 is ≥ 3 . The elementary plaquette trace has four spin-one-half edges and energy 3.

Write $P_0 = |1\rangle\langle 1|$ and $Q_0 = I - P_0$ in the original product Haar space. Integrating one original link gives $P_0 W_p = 0$. Two distinct elementary plaquettes have an edge occurring in just one of their two words; integration over that link vanishes. For $p=q$, the product holonomy has the original Haar law and the fundamental character has squared norm one. Therefore

$$\langle W_p, W_q \rangle_H = \delta_{pq}. \quad (\text{A1})$$

For the complex homogeneous coefficient ζ , $x_p = \zeta$, put $B_\zeta = K - \zeta S$. This complex variable retains the physical return $H = \kappa B_{\xi} + 2\kappa M \xi I$ at real ξ . On C_1 plus $Q_0 H_{\text{phys}}$ the full original block operator is

$$\begin{aligned} B_\zeta &= [[0, Z], [W, D]], \\ D &= Q_0(K - \zeta S)Q_0 \text{ on } \text{Dom}(K) \cap Q_0 H_{\text{phys}}, \\ W &= -\zeta \sum_p W_p, \quad Z = -\zeta P_0 S Q_0. \end{aligned} \quad (\text{A2})$$

Z is the original complex-linear row, with no conjugation of its parameter in the analytic extension. Equation (A1) proves exactly

$$\|W\| = \|Z\| = \sqrt{M} |\zeta|.$$

On $|\zeta| \leq 1/M$, $\text{Re} \langle D h, h \rangle \geq \|h\|^2$ since $\| \zeta S \| \leq 2$. The closed operator D is a bounded perturbation of $Q_0 K Q_0$ on the same domain. For $\text{Re } e < 1$ its inverse $D - e$ exists with norm $\leq (1 - \text{Re } e)^{-1}$: injectivity and closed range follow from the real part inequality; the same inequality for its adjoint gives dense range, hence surjectivity. All finite boxes here have $M = 3(2L)^2(2L+1) \geq 240$.

B.2 A2. A complete ground/complement decomposition on the complex disk

The scalar equation for the small eigenvalue is

$$f(e) = e + Z(D - e)^{-1} W = 0. \quad (\text{A3})$$

On $|e|=2/M$ the second term has modulus at most $1/(M-2)<2/M$. The finite scalar argument principle, or Rouché's theorem applied to e , gives one simple zero inside that circle, with multiplicity counted. The contour construction makes $e(\zeta)$ analytic on a neighborhood of the closed $|\zeta|\leq 1/M$ disk and $|e|\leq 2/M$. Define the actual right and left graph coordinates

$$\begin{aligned} w &= -(D-e)^{-1}W, \\ ell &= -Z(D-e)^{-1}, \\ P &= (1, w)(1, ell)/(1+ell\ w). \end{aligned} \quad (A4)$$

They satisfy $B(1, w)=e(1, w)$, $(1, ell)B=e(1, ell)$, $ell\ W=e$, and $Z+ell\ D=e\ ell$. Their norms obey

$$\begin{aligned} ||w||, ||ell|| &\leq \sqrt{M}/(M-2), \\ M/(M-2)^2 &\leq 240/238^2 < 1/225. \end{aligned}$$

The last bound decreases with $M\geq 240$ by differentiating $M/(M-2)^2$. Consequently $1+ell\ w$ is nonzero, $P^2=P$, and

$$||P||_{\text{trace}} \leq (1+1/225)/(1-1/225)=113/112. \quad (A5)$$

The original complement is $\ker(1, ell)$. Its exact coordinate map and inverse are

$$\begin{aligned} J: Q0H &\rightarrow \ker(1, ell), \quad Jh=(-ell\ h, h), \\ J^{-1} &= Q0 \text{ restricted to } \ker(1, ell). \end{aligned} \quad (A6)$$

On the unchanged operator domain direct multiplication gives

$$(B-e)J=J\ Bc, \quad Bc=D-e-Well. \quad (A7)$$

Every factor in the feedback $Well$ is retained. For its real part,

$$\begin{aligned} \text{Re}\langle Bc\ h, h \rangle &\geq [1-2/M-1/(M-2)]||h||^2 \\ &\geq [1-1/120-1/238]||h||^2 \\ &> (49/50)||h||^2. \end{aligned} \quad (A8)$$

Bc is again the same self-adjoint free operator plus a bounded perturbation. The norm-convergent Duhamel series constructs its strongly continuous semigroup; its derivative on $\text{Dom}(K)$, followed by the real-part energy inequality (A8), proves $||\exp(-\tau Bc)||\leq \exp(-49\tau/50)$. Density extends this estimate to all vectors. Holomorphy in ζ follows locally from the uniformly convergent bounded perturbation series and the analytic coefficients in (A4).

The exact excited semigroup is therefore

$$\exp[-\tau(B-e)](I-P)=J\ \exp(-\tau Bc)Q0(I-P). \quad (A9)$$

The bounds on the original graph maps are

$$\begin{aligned} ||J|| &\leq 226/225, \\ ||Q0P|| &\leq 113/1680, \\ ||Q0(I-P)|| &\leq 1793/1680. \end{aligned} \quad (A10)$$

For example $Q0P=w(1, ell)/(1+ell\ w)$, and using $||w||\leq 1/15$ and $\sqrt{1+1/225}\leq 226/225$ gives $113/1680$. No product-space Gram is replaced.

For real ξ , B_{ξ} is the full self-adjoint physical operator less its displayed scalar. Its lowest eigenvalue is ≤ 0 by the original constant trial state. Equation (A8) leaves the entire remaining spectrum positive after subtracting e . Thus e is its actual ground eigenvalue. P is its actual orthogonal ground projection, obtained above on the same domain. This also identifies the analytic branch without choosing a surrogate vacuum.

B.3 A3. The complete all-time analytic bound

Define the analytic connected heat observable by the literal trace

$$\begin{aligned} \text{Chat_pq}(\tau; \zeta) \\ = \text{Tr}[P \, W_p \exp[-\tau(B-e)] (I-P) W_q]. \end{aligned} \quad (\text{A11})$$

At real ξ this equals the original $C_{\text{pq}}(\tau/\kappa; \xi)$, with its actual vacuum-mean subtraction. The complex trace is its holomorphic continuation; no conjugation of ζ is inserted. Each original multiplication W_p has norm at most 2. Equations (A5), (A9), (A10) give

$$\begin{aligned} |\text{Chat_pq}(\tau; \zeta)| \\ \leq 4(113/112)(226/225)(1793/1680) \exp(-49\tau/50) \\ \leq 5 \exp(-49\tau/50), \quad |\zeta| \leq 1/M, \tau \geq 0. \end{aligned} \quad (\text{A12})$$

The original link-center substitution $U_{(n,i)} \rightarrow (-1)^{\sum_{j<i} n_j} U_{(n,i)}$ sends every plaquette trace to $-W_p$, preserves Haar, K and all gauge spaces, and is its own inverse. It carries B_{ζ} to $B_{-\zeta}$. Both marked traces change sign, so (A11) is even in ζ . Cauchy's coefficient formula on the original circle $|\zeta|=1/M$ and summation of the full even tail now prove

$$\begin{aligned} |\text{Chat_pq}(\tau; \xi) - C_0(\tau) - \xi^2 C_2(\tau) - \xi^4 C_4(\tau)| \\ \leq 5 \exp(-49\tau/50) (M|\xi|)^6 / [1 - (M|\xi|)^2], \\ |\xi| < 1/M, \tau \geq 0. \end{aligned} \quad (\text{A13})$$

All constants and all higher orders remain. The bound is uniform in positive time, with its original volume dependence. It also justifies differentiation in the source before Laplace integration. The finite rational functions in the coefficient proof are therefore precisely the Taylor coefficients of (A11). The Laplace uniqueness argument H3 returns their actual time functions.

B.4 A4. A positive original opposite-face correlation on an entire time interval

For the two faces H17, $C_0=C_2=0$ and $C_4=h_4=4P+B$ with H18--H20. Exact rational Taylor remainder estimates enclose $\exp(-x)$ for each x in $[0,50]$: the even Taylor sum of degree 240 is an upper bound, and subtracting $x^{241}/241!$ is a lower bound. Taylor's Lagrange remainder has the required negative sign and magnitude $\leq x^{241}/241!$. The code checks positive lower endpoints. This is an interval calculation with rational endpoints, including at all cited times.

At $L=2$, $M=240$, $\xi=10^{-10}$, $\tau=1$, evaluation of H18--H20 and (A13) gives

$$85879/10^7 < \text{Chat_pq}(1; \xi)/\xi^4 < 85951/10^7. \quad (\text{A14})$$

The complete rational inner endpoints are in generated/heat_bounds.json. The physical parameters are $g^2=50000$, $\kappa=100000/a$, and $t=1/\kappa$.

The positivity on $1 \leq \tau \leq 3$ has a continuous proof. In the first bracket of H18, let $a=1/882$, $b=-4441/3611790$, $c=75173779/67612708800$. Its quadratic minimum is $c-b^2/(4a)$. Exact rational comparison gives

$$c-b^2/(4a) > (121/108864)/4 + (197/9988160)/16. \quad (\text{A15})$$

Since $\exp(-3/2) < 1/4$ and $\exp(-7/2) < 1/16$, the two negative exponentials in H18 are bounded by this quantity times $\exp(-3\tau)$ for $\tau \geq 1$. Its other terms are positive. Thus $P(\tau) > 0$ throughout $\tau \geq 1$. In H19 the first bracket is at least $5/648$ at $\tau=1$ and increases, and both other terms are positive. Hence

$$h_4(\tau) \geq (5/648)\exp(-3\tau), \quad \tau \geq 1. \quad (\text{A16})$$

For τ in $[1,3]$, the ratio of the complete error (A13) to ξ^4 times (A16) is at most

$$648 M^6 \xi^2 \exp(303/50)/[1-(M\xi)^2] < 0.531 < 1 \quad (\text{A17})$$

at the actual M and ξ above. Consequently the full interacting-vacuum correlation, not just its leading coefficient, is strictly positive on $1/\kappa \leq t \leq 3/\kappa$. Its original integrated value has all physical factors retained by H4.

B.5 A5. Full inverse-power metrics, including the original state norm

Let $R: C^M \rightarrow H_{\text{phys},0}$ have the original centered columns r_p . For integer $k \geq 1$,

$$\begin{aligned} G^-(k) &= \kappa^k R^* A^{(-k)} R \\ &= \int_0^\infty \tau^{(k-1)} \text{Chat}(\tau) d\tau / (k-1)!. \end{aligned} \quad (\text{A18})$$

The actual finite-regulator spectral theorem and the positive centered gap established in A2 justify the integral. Equation (A12) gives a uniform integrable majorant for the analytic Taylor coefficients and their remainder. For a partial fraction $a/(z+c)^n$, its exact contribution is

$$a (k+n-2)! / [(k-1)!(n-1)! c^{(k+n-1)}]. \quad (\text{A19})$$

Every original pole and multiplicity is used. Integrating (A13) proves the entrywise bound

$$\begin{aligned} |G^-(k)_{pq} - \sum_{j=0}^2 \xi^{(2j)} G^-(k)_{(2j),pq}| \\ \leq 5(50/49)^k (M|\xi|)^6 / [1-(M|\xi|)^2]. \end{aligned} \quad (\text{A20})$$

The physical energy response is $G^-(1)$. For the actual derivative primitive

$$\begin{aligned} \Phi &= \kappa A^{(-1)} R, \\ A \Phi &= \kappa R, \quad \Phi * \Phi = G^-(2), \\ \Phi * A \Phi &= \kappa G^-(1), \end{aligned} \quad (\text{A21})$$

the original state and energy metrics remain separately specified.

All coefficients for $k=1,2$ on the original 240-face $L=2$ box are calculated entry by entry in generated/native_inverse_metrics_L2.json. Let $L_-(k,j)$ be the maximum absolute row sum of the exact degree- $2j$ coefficient matrix. A real symmetric M by M error whose entries have magnitude at most e has operator norm at most $M e$: apply the row and column bounds and Cauchy--Schwarz. Therefore (A20), with the complete off-diagonal rows, gives

$$\begin{aligned} [3^{(-k)-w_k} I] &\leq G^-(k) \leq [3^{(-k)+w_k} I], \\ w_k &= \xi^2 L_-(k,1) + \xi^4 L_-(k,2) \\ &\quad + 5M(50/49)^k (M\xi)^6 / [1-(M\xi)^2]. \end{aligned} \quad (\text{A22})$$

For $0 < \xi \leq 1/1600$, all the scalar terms on the right increase with ξ , so its endpoint evaluation bounds the whole interval. Exact rational evaluation gives

$$\begin{aligned} (3/10)I_{240} < G^{\wedge}(1) < (11/30)I_{240}, \\ (9/100)I_{240} < G^{\wedge}(2) < (13/100)I_{240}. \end{aligned} \quad (A23)$$

This is the original $L=2$ box, every $a > 0$, and every $g^2 \geq 20$. The identities use the original coefficient pairing on $C^{\wedge}240$; the actual state Gram is the calculated $G^{\wedge}(2)$, not the identity. The complete original energy form on these primitive columns is $\kappa G^{\wedge}(1)$. The tighter rational endpoints and each original row-sum certificate are retained in generated/heat_bound s.json.

B.6 A6. Scope and comparison with the preceding analysis

The finite-volume Cauchy circle in (A13) has radius $1/M$. Equation (A23) has been evaluated on the original $L=2$ family, with all 240 columns. Every dependence on M , a , g and time is displayed. These statements establish neither a nontrivial four-dimensional continuum field nor a uniform-in-volume positive mass bound. They do not independently recertify the earlier uniform-source/gap argument.

The method uses classical bounded perturbation, block resolvents, the scalar argument principle and the original spectral theorem. A relevant primary Feshbach--Schur antecedent is Dusson--Sigal--Stamm, arXiv:2105.02058. The present proof gives all blocks, graph inverses and finite constants on the actual Hamiltonian instead of supplying a desired spectral gap as an input. The new Split-Zero heat transfer to the resulting actual Grams is in RH_HEAT_TRANSFER.md.

B.7 A7. The original forcing Gram for the same observation family

The time-zero matrix is the original forcing Gram $G^{\wedge}(0) = R^*R = \text{Chat}(0)$. It is computed from exactly the same complete pole coefficients by

$$G^{\wedge}(0)_{(2j),pq} = \sum_{c>0} a_{(c,1)}, \quad (A24)$$

because higher pole orders vanish at $\tau=0$. Equation (A13) at zero supplies its complete entry error $5(M\xi)^6/[1-(M\xi)^2]$. The identical full row-sum argument on the actual $L=2$ box and $0 < \xi \leq 1/1600$ proves

$$(49/50)I_{240} < G^{\wedge}(0) < (51/50)I_{240}. \quad (A25)$$

No division by a chosen source norm is involved. The tighter rational interval is retained with $G^{\wedge}(1), G^{\wedge}(2)$ in generated/heat_bounds.json. Thus all three original Grams R^*R , $\Phi^*A\Phi/\kappa$ and $\Phi^*\Phi$ are quantitatively controlled together. Their use in the actual inverse-power observation is proved in T7 of the source-transfer note.

B.8 A8. The full original kinetic Gram, retaining its local zero entries

The forcing columns have original energy Gram $R^*A R = \kappa Kobs$, where

$$Kobs_{pq} = \langle \Gamma(w_p, w_q) \rangle_{\rho} = -\text{Chat}'_{pq}(0). \quad (A26)$$

The ground-state form identity proves the first equality; differentiation of the original semigroup proves the second. In the coefficient file each simple pole contributes its energy times its coefficient and each double pole contributes minus its coefficient. Higher poles contribute zero. This gives all entries through degree four in `native__inverse__metrics__L2.json`.

Every off-diagonal entry outside the original face adjacency is zero exactly: Γ differentiates a common original edge. On a diagonal, $\Gamma(W_p, W_p) = 4 - W_p^2$ has absolute value at most 4. On adjacent distinct faces, their single shared-edge derivative vectors each have Euclidean length at most one; writing the original SU(2) product as $q_0 I - i q \cdot \sigma$ gives the length $\sqrt{1 - q_0^2}$. Their scalar product has absolute value at most 1. Each face has at most 12 original neighbours. All assertions concern the unchanged original link derivatives, with their generator factor $1/2$ retained.

The complex ground projection from (A4)–(A5) therefore bounds the diagonal expectation by $4 \cdot 113/112$ and each adjacent expectation by $113/112$. Original center symmetry makes these analytic expectations even. Cauchy's formula on $|\zeta| = 1/M$ bounds the entire row of the omitted tail by

$$(113/112) \cdot 16 \cdot (M|x_i|)^6 / [1 - (M|x_i|)^2]. \quad (\text{A27})$$

This uses the exact local support, rather than a dense- M multiplier on this particular Gram. The exact degree-two and degree-four row sums and (A27) give, on the original $L=2$ box with $g^2 \geq 20$,

$$(29/10) I_{240} < K_{\text{obs}} < (31/10) I_{240}. \quad (\text{A28})$$

The sharper rational endpoints and every nonzero coefficient are retained in the same records. This is the further original energy input used in T8 to bound the correction between the actual state-minimum and energy-minimum sections of the inverse-power family.

C The Split-Zero heat comparison transferred to the original Yang-Mills Grams

21 September 2026. This is a source-specific transfer, accompanied by a new calculation of actual Yang-Mills heat correlations. The inspected Riemann workbench revision is 128aa308dd2a70ba6e073816a957d1ee6414ba2c. The exact read ranges and source blobs are in SOURCE_INTAKE.json. Its arithmetic quantities remain attached to their own source; the Yang-Mills quantities below are computed from the original Hamiltonian and its physical vacuum.

C.1 T1. What was read and what is transferred

The complete argument used from the Riemann workbench is HM6--HM14 and HM19--HM24 of HEAT_TO_ORIGINAL_QUOTIENT_METRICS.tex (19 September). It constructs its original coefficient column map R_{cal} and a finite heat replacement $R_{\text{cal}}_J = R_{\text{cal}} - T_s P_{\text{cal}}$. The physical arithmetic source pairing is unchanged and $T_s^* T_s = s^{-1}I$. From the complete original Gram, including both relation cross blocks, HM13 proves the relative column error. Triangle inequalities give $(1-\delta)^2 G \leq G_J \leq (1+\delta)^2 G$. Completing the square on the same coefficient fibers carries these bounds through their actual restrictions and quotient minima. HM22--HM24 retain every determinant rank and give the finite heat depth for a prescribed return error.

HG6--HG10 of ORIGINAL_HEAT_AND_SOURCE_ENDPOINT.tex were also read. They expand actual observation/kernel currents with both mixed products and give their error under a relative change of the same original Gram. The fixed-divisor subexponential estimate HG1--HG5 is retained as an inspected arithmetic result; no value of its divisor-dependent constants is assigned to a Yang-Mills bound.

The common column-map calculation is instantiated below, on actual Yang-Mills spaces. Its input error is supplied by the original physical semigroup rather than by the arithmetic dilation. A source-specific map between arithmetic zeta packets and physical gauge fields is not asserted. The explicit shared interface and each new map are the displayed linear maps below.

C.2 T2. The original physical heat columns and their entire forcing defect

Fix the original $L=2$ open box, $M=240$, $a>0$, and $g^2 \geq 20$. Let $A=H-E_0$ on the centered physical Hilbert space H_0 , with its unchanged inner product. The source columns R have entries $rp = (Wp - \langle Wp \rangle_{\rho})\psi$. From A23,

$$\begin{aligned} \Phi &= \kappa A^{(-1)}R, \\ G_2 &= \Phi * \Phi, \quad E = \Phi * A \quad \Phi = \kappa G_1, \\ (9/100)I &< G_2 < (13/100)I, \\ (3\kappa/10)I &< E < (11\kappa/30)I. \end{aligned} \tag{T1}$$

All original coefficient axes p are retained. In particular Φ is injective. The finite-regulator inverse and the original source identity $A \Phi = \kappa R$ are established in the analytic proof, not assumed as an observation metric.

The physical gap needed here has an independent short derivation. In the original Haar space, minmax applied to $B_{\xi} = K_{\xi} S$ gives its second physical eigenvalue $\geq 3-2M_{\xi}$. Its ground eigenvalue is ≤ 0 by the original constant trial vector. Therefore, for the actual centered A ,

$$A \geq \kappa(3-2M_{\xi})I \geq (27\kappa/10)I \tag{T2}$$

because $\xi \leq 1/1600$. This is a bound for this original box and coupling domain. Every volume factor in its derivation is explicit.

For $T > 0$ define the actual map and its convergent inverse on H_0

$$\begin{aligned} C_T &= I - \exp(-TA), \\ C_T^{-1} &= \sum_{j \geq 0} \exp(-jTA), \\ \|C_T^{-1}\| &\leq [1 - \exp(-27\kappa T/10)]^{-1}. \end{aligned} \quad (T3)$$

They commute with A on $\text{Dom}(A)$; convergence in the graph norm follows by commuting A through each term and applying the same geometric bound. Put

$$\begin{aligned} \Phi_T &= C_T \Phi, \quad \kappa \int_0^T \exp(-tA) R \, dt, \\ E_T &= \Phi_T^* A \Phi_T, \quad G_2_T = \Phi_T^* \Phi_T. \end{aligned} \quad (T4)$$

Both state and energy errors retain their complete mixed products. Specifically,

$$\begin{aligned} E_T - E &= -\Phi^* A \exp(-TA) \Phi \\ &\quad - \Phi^* \exp(-TA) A \Phi \\ &\quad + \Phi^* \exp(-TA) A \exp(-TA) \Phi, \end{aligned} \quad (T5)$$

with the analogous identity omitting A for G_2 . The two cross products are equal here because the original A and its heat semigroup commute; their origin in the expansion remains explicit. The source equation is

$$A \Phi_T = \kappa R - \kappa \exp(-TA) R. \quad (T6)$$

Thus the original forcing defect and its primitive are exactly $-\kappa \exp(-TA) R$ and $-\exp(-TA) \Phi$, respectively.

For $\epsilon = \exp(-27\kappa T/10)$, spectral calculus applied on the complete centered physical space gives

$$\begin{aligned} (1-\epsilon)^2 G_2 &\leq G_2_T \leq G_2, \\ (1-\epsilon)^2 E &\leq E_T \leq E. \end{aligned} \quad (T7)$$

For the energy inequality one applies C_T to $A^{1/2} \Phi$; for the state inequality one applies it to Φ . These are the actual column-map instances of HM11, with a sharper one-sided upper bound supplied by the semigroup. The norm of the omitted column is bounded relatively in its own original state/energy pairing. All constants are independent of A after the exact physical time T is inserted; T itself retains its factor $1/\kappa$.

C.3 T3. Original support quotients and the corrected minimum

For an original face subset F , put $V_F = \Phi(C^F)$, $U_F = A V_F = \kappa R(C^F)$. The complex is

$$V_F \xrightarrow{-A} H_0 \xrightarrow{-0} 0. \quad (T8)$$

Its support transition $F \subset G$ is the actual inclusion in degree zero and identity on H_0 . The differential square commutes by the same original A . For $J = G \setminus F$, the transported cohomology kernel is exactly

$$\begin{aligned} V_G/V_F &\rightarrow \ker[H_0/U_F \rightarrow H_0/U_G], \\ [h] &\rightarrow [Ah]. \end{aligned} \quad (T9)$$

Surjectivity follows by writing a killed class as Ah with h in V_G . Changing h by V_F changes Ah by U_F , and conversely the injectivity of A on H_0 implies that a change by U_F has exactly a change by V_F . The inverse is $[Ah] \rightarrow [h]$. The heat complexes use $V_F \hat{T} = C_{TV_F}$ with the same target H_0 and differential A . The actual pair (C_T, C_T) intertwines their differentials and is invertible by (T3). It commutes with every support inclusion. Applying the Split-Zero support reconstruction therefore retains F and its receiving zero, together with the original killed representative and its primitive.

The energy-minimum representative of $[\Phi_J y]$ in V_G/V_F has the old-support coordinate

$$x_F = -E_{FF}(-1)E_{FJ} y.$$

All blocks are taken in the fixed original face coordinates. Its exact quotient Gram is

$$Q_E(F, G) = E_{JJ} - E_{JF} E_{FF}(-1)E_{FJ}. \quad (T10)$$

Completing the square in x_F proves this formula and both inverse coordinate maps to the quotient. The removed primitive $\Phi_F x_F$ remains. The state minimum has the separately calculated formula with G_2 ; it is not substituted for the energy minimum.

Equation (T7) holds at every vector (x_F, y) in the same original coefficient fiber. Taking the infimum in x_F on both sides proves

$$(1-\epsilon)^2 Q_E(F, G) \leq Q_{-T}(F, G) \leq Q_E(F, G), \quad (T11)$$

and the state counterpart. The same proof applies successively to every finite chain of fixed restrictions and quotient maps, retaining the same two bounds, not multiplying an error factor at each stage. This is the actual HM19--HM21 minimum-fiber argument in the physical metric. A rank-zero quotient retains its unique zero vector and determinant one.

To retain the surrounding physical space, put $P_\Phi = \Phi G_2(-1)\Phi^*$. Its multiplication identities give $P_\Phi^2 = P_\Phi = P_\Phi^*$. For any original form-domain h perpendicular to $\text{im}(\Phi)$, the full energy is

$$\begin{aligned} q_A(\Phi c + h) &= \kappa c^* G_1 c \\ &\quad + 2\kappa \text{Re}\langle R c, h \rangle + q_A(h). \end{aligned} \quad (T12)$$

This is the complete original coupling to the remaining physical functions. No reducing-subspace property of the finite observation family is used.

C.4 T4. A finite heat horizon for every four-return determinant

For a rank- r positive form G and its same-coordinate heat comparison satisfying (T7), its relative eigenvalues belong to $[(1-\epsilon)^2, 1]$. The determinant ratio is invariant under the displayed congruence with $G^{(-1/2)}$, including its inverse $G^{(1/2)}$. Hence

$$|\log \det G_T - \log \det G| \leq 2r[-\log(1-\epsilon)]. \quad (T13)$$

The difference uses the same original frame and, for energy Grams, the same κ^r factor, which cancels exactly in the ratio. For any four prescribed restriction/quotient returns with coefficients +1,+1,-1,-1 and ranks $r_j \leq 240$,

$$|L_T - L| \leq 2 \sum_j r_j [-\log(1-\epsilon)] \leq 1920 \epsilon / (1-\epsilon). \quad (T14)$$

The final inequality follows by integrating $1/(1-x)$ from zero to ϵ . Take the actual physical horizon

$$T = 10/\kappa = 5a/g^2. \quad (T15)$$

Then $\epsilon = \exp(-27)$. The exact rational exponential enclosure gives

$$\epsilon < 19/10^{13}, \quad 1920 \epsilon / (1-\epsilon) < 4/10^9. \quad (T16)$$

The calculated inner upper bound is approximately $3.608695327762 \cdot 10^{-9}$. The complete rational fraction is in generated/heat_bounds.json. This is one common finite heat horizon for all four original returns and all original support choices in this 240-column family. It requires no estimate of a newly selected arithmetic comparison constant or observation angle.

C.5 T5. Every signed observation current retains its mixed terms

Fix a surjection Λ from the original C^{240} onto its actual chosen observation image; for example its rows can read a fixed subset of original face coordinates. From the energy Gram E define

$$\begin{aligned} Q &= (\Lambda E^{-1} \Lambda^*)^{-1}, \\ \Omega &= \Lambda^* Q \Lambda, \\ L &= E - \Omega. \end{aligned} \quad (T17)$$

The minimum representative $S = E^{-1} \Lambda^* Q$ satisfies $\Lambda S = I$ and $E S = \Lambda^* Q$. Direct expansion proves that $y^* L y$ is the squared E -norm of $(1 - S \Lambda) y$. Thus both the kernel residual and boundary Gram are the original canonical ones, with every rectangular mixed entry preserved. Define the heat versions using E_T and the identical Λ .

Put $\eta = 2\epsilon - \epsilon^2$. Equation (T7) gives $(1-\eta)E \leq E_T \leq E$. Minimize on the identical Λ fibers to get the same bounds for Q . For fixed original columns a_1, a_2, v and $\omega > 0$ let

$$\begin{aligned} K_i &= a_i^* L v, \quad B_i = a_i^* \Omega v, \\ d_i &= a_i^* E a_i, \quad E_v = v^* E v, \\ \Phi_K &= 2\operatorname{Re}(\operatorname{conj}(K_1) K_2) / \omega, \\ \Phi_B &= 2\operatorname{Re}(\operatorname{conj}(B_1) B_2) / \omega, \\ \Phi_{\text{cross}} &= 2\operatorname{Re}(\operatorname{conj}(K_1) B_2 + \operatorname{conj}(B_1) K_2) / \omega. \end{aligned} \quad (T18)$$

The same-fiber bounds and Cauchy--Schwarz give $|\Delta K_i| \leq \eta \sqrt{d_i E_v}$, $|\Delta B_i| \leq \eta \sqrt{d_i E_v}$, while each original $|K_i|$ and $|B_i|$ is at most $\sqrt{d_i E_v}$. Expanding every linear and quadratic product in (T18) proves the HG8 receiving inequalities

$$\begin{aligned}
|\Delta \Phi_K| &\leq 2\sqrt{d_1 d_2} E_v(4\eta + 4\eta^2)/\omega, \\
|\Delta \Phi_B| &\leq 2\sqrt{d_1 d_2} E_v(2\eta + \eta^2)/\omega, \\
|\Delta \Phi_{\text{cross}}| &\leq 2\sqrt{d_1 d_2} E_v(6\eta + 4\eta^2)/\omega. \quad (\text{T19})
\end{aligned}$$

These estimates contain the original phases and both cross products. The actual source/target numerical values in this application are the calculated Yang–Mills Grams, and their relative error is (T16). No sign of a current is inferred from its norm alone.

C.6 T6. Source and execution scope

This contribution reads the original HM/HG heat and metric proofs and proves the displayed Yang–Mills instances in full. The arithmetic dilation T_s , its Mellin variables, zeta-zero jets, original source masses and period-dependent constants remain in the cited RH construction. Here $\exp(-TA)$ is the physical ground-relative semigroup, with T (T15) in the original time units.

The heat coefficients, original box matrices, graph constants and rational bounds are independently exercised by the executable tests described in VERIFICATION.md. Their complete analytic proofs are supplied as written arguments for review. The tests do not give a Lean proof, independently audit every older source estimate, or establish the four-dimensional continuum gap. The standing continuum path and the growing-volume problem still require bounds beyond the explicitly finite M -dependent domain proved here.

C.7 T7. The latest inverse-power observation has a concrete receiving instance

After the heat-source intake, the complete 21 September source INVERSE_POWER_CONDUCTOR_OBSERVABILITY.tex was read at the same pinned commit. Its IK9–IK10 construct a literal left inverse of selected original conductor rows on their inverse-power space. IK12 returns that inverse through the original source and observation Grams, retaining their eigenvalue factors. IK13 explicitly permits its lower bound to tend to zero with its cutoff. Those coefficients, root differences and guards remain in that construction.

In the present Yang–Mills family the actual inverse-power source is

$$\begin{aligned}
\Phi &= \kappa A^{(-1)} R : C^{240} \rightarrow H_0, \\
0 &= R^* : H_0 \rightarrow C^{240}, \\
0 \Phi &= G^{(1)}. \quad (\text{T20})
\end{aligned}$$

The independent computed bound (A23) makes this original response matrix invertible, so its exact coefficient left inverse is

$$\begin{aligned}
W &= (G^{(1)})^{(-1)} R^*, \\
W \Phi &= I_{240}, \quad ||(G^{(1)})^{(-1)}|| \leq 10/3. \quad (\text{T21})
\end{aligned}$$

This is a proved inverse on these specific columns, not an assumed observation angle. The original quotient metric for the observation O is $(G^{(0)})^{(-1)}$: minimizing the physical Hilbert norm over $R^*h=z$ has the representative $R(G^{(0)})^{(-1)}z$. Its full residual is perpendicular to $\text{im}(R)$, as multiplication by R^* verifies. Thus $P_R = R(G^{(0)})^{(-1)}R^*$ is the actual orthogonal projection, and every nonzero coefficient vector c satisfies

$$\begin{aligned}
||P_R \Phi c||^2 &= c^* G^{(1)} (G^{(0)})^{(-1)} G^{(1)} c, \\
||\Phi c||^2 &= c^* G^{(2)} c. \quad (\text{T22})
\end{aligned}$$

Returning the three computed original metrics (A23),(A25) gives

$$\begin{aligned} G^-(1)(G^-(0))^{-1}G^-(1) &\geq (3/34)I, \\ ||P_R \Phi c||^2 / ||\Phi c||^2 &> 150/221. \end{aligned} \quad (T23)$$

Indeed the first coefficient is $(3/10)^2/(51/50)=3/34$, and dividing by the upper state coefficient $13/100$ gives $150/221$. Every source metric and its inverse has been retained. This is the IK12 left-inverse/minimum argument applied to the original physical inverse-power family, with the needed Gram bounds actually calculated here. It detects all 240 columns simultaneously at $L=2$, every $a>0$ and $g^2 \geq 20$. It does not supply an infinite-volume fraction or an independence assertion for an uncomputed concatenation of higher powers.

The portion outside this observation is the exact vector $(I-P_R)\Phi c$. Its squared norm is

$$c*[G^-(2)-G^-(1)(G^-(0))^{-1}G^-(1)]c, \quad (T24)$$

and is positive semidefinite by the original orthogonal decomposition. No component is removed from the energy formula (T12). The maps (T20)–(T24) state the entire receiving construction; they make no identification of arithmetic zeta zeros with physical spectral points.

C.8 T8. The actual residual between the state and energy sections is bounded

Retain the original orthogonal state projection P_R from T7 and define

$$\begin{aligned} h &= (I-P_R)\Phi c, \\ B_0 &= (G^-(0))^{-1}G^-(1), \quad P_R \Phi c = R B_0. \end{aligned} \quad (T25)$$

All columns of h satisfy $R^*h=0$. Since $A \Phi c = \kappa R$, the mixed energy $q_A(\Phi c, h d)$ vanishes for every coefficient pair c, d , by the original Hilbert adjoint identity. Thus Φc is the actual minimum-energy section of the affine fibers $R^*f = G^-(1)c$. The state-minimum section of that same fiber is $R B_0 c$. They have the exact comparison

$$\begin{aligned} h^*Ah &= \kappa [G^-(1)(G^-(0))^{-1}K_{obs}(G^-(0))^{-1}G^-(1)-G^-(1)], \\ (P_R \Phi c)^*A(P_R \Phi c) &= \Phi c^*A \Phi c + h^*Ah. \end{aligned} \quad (T26)$$

To verify the first identity, expand the whole expression $(\Phi c - R B_0)^*A(\Phi c - R B_0)$. Both cross products are $\kappa G^-(1)$, while the last term is $\kappa B_0^* K_{obs} B_0$; their signed sum is (T26). This also proves nonnegativity of the right side. The state residual remains the matrix (T24).

The three raw bounds (A23),(A25),(A28) give the strict original energy bound

$$0 \leq h^*Ah < (1322/7203) \Phi c^*A \Phi c. \quad (T27)$$

Indeed congruence by $(G^-(1))^{-1/2}$, with its explicitly inverse square root, puts the bracket in (T26) below

$$[(31/10)(50/49)^2(11/30)-1]I = (1322/7203)I.$$

Returning that congruence proves (T27) in the original coefficient coordinates. Thus the state-section correction has at most this fraction (less than 0.184) of the original primitive's energy, and (T23) retains more than 150/221 of its state norm in the actual observed force span. The difference itself remains an element of $\ker(R^*)$ with its complete state and energy Grams, rather

than being dropped. Both conclusions hold simultaneously for all 240 original columns on the specified finite box and coupling domain.

For a prescribed observation z , the two exact sections are $\Phi(\hat{G}(1))^{(-1)}z$ and $R(\hat{G}(0))^{(-1)}z$, and their difference is $h(\hat{G}(1))^{(-1)}z$. Their source-to-observation compositions are both the identity. This supplies the explicit section-correction morphism and its quantitative energy return in the original metric. It is the same minimum-section mechanism read in the Split-Zero source, now with all physical Grams evaluated and the entire correction bounded on its stated domain.

D Original Yang–Mills heat: a box-independent analytic radius and row-sum remainder

21 September 2026. This continuation addresses the two quantities left by the heat edition: its radius $1/M$ and the coupling of its observed states to the full physical complement. All operator coefficients, Haar masses, original link and plaquette coordinates, and state and energy pairings are retained. This note proves the local-source-to-heat argument used here in full. It does not certify every earlier stronger-coupling assertion. Its coefficient tables are the inherited, separately replayed heat tables; new constant and matrix calculations have their own records. No continuum mass-gap conclusion, historical-priority determination, or external analytical review is asserted.

D.1 U1. Original operator, physical time, and coefficient maps

For each $L \geq 2$ take vertices $\{-L, \dots, L\}^3$, all contained positive nearest-neighbor links $e=(n,i)$, and all elementary faces $p=(n;i,j)$, $i < j$. Retain

$$\begin{aligned} W_p &= \text{tr}(U_i(n) U_j(n+e_i) U_i(n+e_j)^{-1} U_j(n)^{-1}), \\ T_a &= -i \sigma_a / 2, \quad X_{e,a} f = d/dt f(\dots, \exp(t T_a) U_e, \dots) | 0, \\ K &= -\sum_{(e,a)} X_{e,a}^2, \\ H(x) &= \kappa K + \kappa \sum_p x_p (2 - W_p), \\ \kappa &= 2g^2/a, \quad x_p = xi/(4g^4), \quad a, g > 0. \end{aligned} \tag{U1}$$

The physical Hilbert space is the invariant subspace of original product Haar probability $L^2(dU)$, with every vertex including boundary vertices averaged. H has physical H^2 domain and H^1 form domain, since the original potential is smooth bounded on this finite compact group. Its smooth positive unit ground vector ψ and simple bottom energy E_0 follow from the modulus form inequality, elliptic regularity, the maximum principle, and the identity

$$q_-(H-E_0)(\psi f) = \kappa \int \psi^2 \sum_i |X_i f|^2. \tag{U2}$$

The multiplication map $f \rightarrow \psi f$ and inverse division by ψ are unitary between $L^2(\rho dU)$ and original $L^2(dU)$, $\rho = \psi^2$. The time coordinates are $\tau = \kappa t$ and $t = \tau/\kappa$. No field variable is rescaled.

For a product-spin label $j=(j_e)$, define $\pi_j(U) = \text{tensor}_e \pi_{j_e}(U_e)$, $d_j = \text{product}_e (2j_e + 1)$, $c_j = \sum_e j_e(j_e + 1)$, and the actual coefficient

$$\begin{aligned} A_j &= d_j \int f(U) \pi_j(U)^* dU, \\ f(U) &= \sum_j \text{Tr}(A_j \pi_j(U)). \end{aligned} \tag{U3}$$

This Fourier map and its inverse follow from the original Haar matrix-coefficient orthogonality. The trace-coefficient norm is $\|f\|_X = \sum_j \|A_j\|_1$. Let X_0 be its zero-Haar physical part, and Y_0 its dense domain $\sum_{(j \neq 0)} c_j \|A_j\|_1 < \infty$. The actual K maps Y_0 bijectively onto X_0 by multiplying each coefficient by c_j ; its inverse divides it by that same c_j . These are auxiliary convergence estimates on the original coefficients, not replacements for (U2).

The compact-group Fourier algebra and its product norm are classical (Eymard, 1964). Here their required inequalities are proved directly. In a product of two coefficient functions, decompose $\pi_j \text{ tensor } \pi_k$ into its full irreducible and multiplicity spaces by a unitary intertwiner. Pinching to all its diagonal representation blocks and tracing each multiplicity space returns the exact output coefficients. Pinching is an average of unitary conjugations. For a partial trace, trace-norm duality gives $|\text{Tr}[(Z \text{ tensor } I)C]| \leq \|C\|_1$ when $\|Z\| \leq 1$. Hence

$$\begin{aligned} ||fh||_X &\leq ||f||_X ||h||_X, \\ ||A(X_e, a f_j)||_1 &\leq j_e ||A_j||_1. \end{aligned} \quad (U4)$$

Every representation dimension and every multiplicity is included in this argument.

D.2 U2. The graph bound that enters the source estimate

A nonzero gauge-invariant coefficient has an invariant tensor on the original incident indices at every vertex. Grouping the factors gives an intertwiner from the dual spin- j_e factor into the tensor product of the others. The maximum target weight is their summed spin, so

$$j_e \leq \sum_{(f \text{ incident } v, f \neq e)} j_f. \quad (U5)$$

Fix $e = \{u, v\}$, and let $E1$ be the other edges at its endpoints. Because the graph is simple and triangle-free, their outer endpoints are distinct. Put $J1 = \sum_{(f \text{ in } E1)} j_f$. The two endpoint inequalities give $J1 \geq 2j_e$. Sum the outer-endpoint inequalities: every edge outside $E1$ and $\{e\}$ is counted at most twice, so $J1 \leq 2J2$, where $J2$ counts these outer edges once. Therefore $\sum_f j_f \geq j_e + J1 + J2 \geq 4j_e$. The original half-integer spectrum obeys $j(j+1) \geq (3/2)j$, and consequently

$$\begin{aligned} c_j &\geq 6j_e \text{ on every physical block;} \\ \sum_e j_e &\leq (2/3)c_j \text{ on every product block.} \end{aligned} \quad (U6)$$

In particular $c_j \geq 3$ on each nonconstant physical block. The fundamental elementary plaquette has $c_j = 3$. This argument includes active bridges and boundary vertices; no cycle decomposition of every active edge is presumed.

D.3 U3. Local labelled source and its exact assembly kernel

For every nonempty finite edge label S retain a zero-Haar physical coefficient family $A_-(S, j)$, with $\text{supp}(j)$ contained in S ; the label may exceed the active support. Put

$$||v||_{\text{loc}} = \sup_e \sum_{(S \text{ containing } e, j \neq 0)} c_j ||A_-(S, j)||_1. \quad (U7)$$

On a finite graph this is a complete space, and its total c -weighted coefficient sum is at most $|E_L| ||v||_{\text{loc}}$. Assembly sends $A_-(S, j)$ to $\sum_{(S \text{ containing } \text{supp}(j))} A_-(S, j)$. Its kernel is exactly the family equations that all those sums vanish. For each S strictly containing $\text{supp}(j)$, the relation with coefficient A at (S, j) and $-A$ at $(\text{supp}(j), j)$ has zero assembly. The actual coefficients at all enlarged S give a unique decomposition of every assembly-kernel element into these relations. Its inverse reads those same enlarged-label coefficients. This supplies the support-preserving quotient map and both inverse laws.

Define, retaining the original union label,

$$\begin{aligned} B(v, h)_S &= K^{-1} Q_H \sum_{(S1 \text{ union } S2=S)} \Gamma(v_{S1}, h_{S2}), \\ \Gamma(f, h) &= \sum_{(e, a)} (X_e, a f) (X_e, a h), \quad Q_H = I - P_H. \end{aligned} \quad (U8)$$

Each Haar scalar removed by Q_H is recorded separately. For an anchor in $S1$, (U4) and (U6) bound the output before K inversion by

$$\begin{aligned}
& 3 \sum_{\text{S1 containing anchor}, j} ||A_{\text{S1}, j}||_1 \\
& \quad \sum_e j_e \sum_{\text{S2 containing } e, k} k_e ||B_{\text{S2}, k}||_1 \\
& \leq (1/3) ||v||_{\text{loc}} ||h||_{\text{loc}}.
\end{aligned}$$

The anchor-in-S2 contribution gives the other 1/3. The original output Casimir cancels precisely against its nonzero inverse Casimir in (U8). Thus

$$||B(v, h)||_{\text{loc}} \leq (2/3) ||v||_{\text{loc}} ||h||_{\text{loc}}. \quad (\text{U9})$$

D.4 U4. Exact elementary coefficient and convergent complex source

For the original four-link word, the coefficient matrix on $(C2)^{\wedge \text{tensor4}}$ has entries

$$\begin{aligned}
& A_{(i1, i2, 1-i2, 1-i3 ; i0, i1, 1-i3, 1-i0)} = (-1)^{(i0+i2)}, \\
& i0, i1, i2, i3 \text{ in } \{0, 1\}.
\end{aligned} \quad (\text{U10})$$

The identity $(U^{(-1)})_{\text{--}}(r, c) = (-1)^{(r+c)} U_{\text{--}}(1-c, 1-r)$ proves its trace formula. Its four nonzero 2-by-2 blocks, up to row/column order, are $[[1, -1], [-1, 1]]$. Direct multiplication gives $(A^* A)^2 = 4A^* A$ and $\text{Tr}(A^* A) = 16$. Thus $|A| = A^* A / 2$ and $||Wp||_{\text{--}}X = 8$. At most four original plaquettes meet an edge.

Allow independent complex x_p with $\max_p |x_p| \leq r$. The source equation is

$$v = (1/3) \sum_p x_p Wp + B(v, v). \quad (\text{U11})$$

Its homogeneous order- n coefficient obeys the original binary recurrence, giving

$$\begin{aligned}
& a_n(r) = \text{Cat}_{\text{--}}(n-1) (2/3)^{(n-1)} (32r)^n, \\
& \sum_{n \geq 1} a_n(r) = r_v(r) = (3/4) (1 - \sqrt{1 - 256r/3}).
\end{aligned} \quad (\text{U12})$$

The radius used throughout the new bound is exactly

$$R0 = 3/256. \quad (\text{U13})$$

For $r < R0$ the series is norm-analytic. It converges absolutely also at $r = R0$, where its tail is $(3/4) \text{binom}(2N, N) / 4^N \leq 3/(4\sqrt{N+1})$. The finite coefficient proof is $\text{Cat}_{\text{--}}N / 4^N = 2(b_{\text{--}}N - b_{\text{--}}(N+1))$, $b_{\text{--}}N = \text{binom}(2N, N) / 4^N$. No strict contraction at the boundary is used.

Each order- n label is an edge-connected union of at most n plaquettes. The coefficients agree literally in every original box containing that union. This follows inductively from (U8): derivatives require a shared original edge, and K inversion acts on the same coefficients.

On a finite graph the norm convergence controls every first and second coordinate derivative, since $j_e j_f \leq c_j$. Thus the actual assembled function solves

$$\begin{aligned}
& Kv = \sum_p x_p Wp + \text{Gamma}(v, v) - C(x), \\
& C(x) = P_{\text{--}}H \text{Gamma}(v, v).
\end{aligned} \quad (\text{U14})$$

For real x , $\psi = \exp(v) / \sqrt{\int \exp(2v)}$ is positive and solves the original eigen-equation with

$$E0 = \kappa [2 \sum_p x_p - C(x)]. \quad (\text{U15})$$

Equation (U2) then proves it is the actual ground vector and fixes every scalar. Elliptic bootstrapping makes it smooth. For complex x the nonzero function $\exp(v)$ is still a smooth solution of that eigen-equation. Its squared integral is addressed below, rather than presumed nonzero.

D.5 U5. The full drift, including its scalar row

Set $D_v f = 2 \Gamma(v, f)$. Equations (U4), (U6) give, on the original coefficient spaces,

$$\begin{aligned} \|D_v h\|_X &\leq \chi(r) \|Kh\|_X, \\ \chi(r) &= (2/3) r_v(r) = (1 - \sqrt{1 - 256r/3})/2 \leq 1/2. \end{aligned} \quad (U16)$$

Both the constant and nonconstant outputs are included. Indeed the sum over generators is at most $6 \sum_j \|A_j(h)\|_1 \sum_{e \in [r_v/6]} [r_v/6]$, and $\sum_{e \in [r_v/6]} \leq (2/3) c_j$. Define

$$\begin{aligned} T &= Q_H D_v K^{-1} : X_0 \rightarrow X_0, \\ p &= P_H D_v K^{-1} : X_0 \rightarrow C, \\ S &= (I - T)^{-1} = \sum_{n \geq 0} T^n. \end{aligned} \quad (U17)$$

The stacked bound is $\|py\| + \|Ty\|_X \leq \chi \|y\|_X$; in particular $\|S\| \leq 1/(1 - \chi)$. Define the actual mean functional by

$$\mu_x(F) = P_{HF} + p S Q_{HF}. \quad (U18)$$

For $\chi \leq 1/2$, $|\mu_x(F)| \leq \|F\|_X$ and $|\mu_x(Q_{HF})| \leq \chi \|Q_{HF}\|_X / (1 - \chi)$. The associated source inverse is $h = K^{-1} S Q_{HF}$ in Y_0 , with

$$(K - D_v)h = F - \mu_x(F). \quad (U19)$$

All maps are analytic for $r < R_0$.

Here (U18) equals the actual complex density functional as well. Put $Z_x = \int \exp(2v)$. Integration by parts gives $\int \exp(2v) (K - D_v)h = 0$ for h in Y_0 , since its first and second derivatives converge. Equation (U19) therefore gives

$$\int \exp(2v) F = \mu_x(F) Z_x. \quad (U20)$$

If Z_x vanished, the left side would vanish for every physical Fourier polynomial. Their uniform closure contains the conjugate of the nonzero smooth gauge-invariant $\exp(2v)$. Its squared modulus would then have zero integral, a contradiction. Hence Z_x is nonzero, and $\mu_x(F) = \int \exp(2v) F / Z_x$. For real x it is precisely expectation in ρ . The exact bilinear integration identity is

$$\mu_x((K - D_v)h \cdot G) = \mu_x(\Gamma(h, G)). \quad (U21)$$

Complex x uses the bilinear analytic continuation; the physical real-coupling Hilbert pairing retains conjugation in its first entry.

D.6 U6. Closed generator and heat flow in the same coefficients

Let $L_x = K - Q_H D_v$ on X_0 , $\text{Dom}(L_x) = Y_0$. The graph norms of K and L_x are equivalent because $\|Q_H D_v h\| \leq \chi \|Kh\|$, $\chi < 1$. Thus L_x is closed and densely defined. For $h > 0$,

$$I + hL_x = [I - hQ_H D_v (I + hK)^{-1}] (I + hK), \quad (U22)$$

and the square bracket is invertible by a Neumann series of norm ratio at most χ . The original sum of trace norms also gives

$$\begin{aligned} & ||(I+hL_x)f||_X \\ & \geq ||f||_X + h(1-\chi) ||Kf||_X \\ & \geq [1+3h(1-\chi)] ||f||_X. \end{aligned} \quad (U23)$$

All powers of the actual resolvent obey the corresponding product bound. The Hille–Yosida/Lumer–Phillips generation theorem applies to this explicitly closed dense domain and the proved surjectivity (U22). It gives

$$\begin{aligned} E_x(\tau) &= \exp(-\tau L_x), \\ ||E_x(\tau)|| &\leq \exp[-d(r)\tau], \quad d(r) = 3(1-\chi(r)) \geq 3/2. \end{aligned} \quad (U24)$$

For completeness, the shifted operator $L_x - d(r)$ is dissipative with the negative-generator convention: the directional norm derivative of $-Kf$ is $-||Kf||$, and its perturbation is at most $\chi ||Kf||$. Its sufficiently large positive resolvent is obtained by (U22) and the resolvent identity. This proves the decay constant, rather than using a norm for one inverse as a substitute for a semi-group estimate. Lumer–Phillips (1961), Theorem 3.1, pp686–687, is the primary generation reference.

Each resolvent in (U22) is analytic in x . Its exponential Euler powers are uniformly bounded on compact complex source polydisks and converge strongly to (U24). Here the latter assertion has a direct integral proof: $(I+hL)^{-1}f = \int_0^\infty \exp(-u)E_x(hu)f \, du$, by integration on the original generator domain and density. The m -fold product at $h=\tau/m$ is the average of $E_x(\tau Y_m/m)f$, where Y_m has the convolution density $\exp(-u)u^{m-1}/(m-1)!$, mean m and variance m . The nonnegative scalar density has mass one. Chebyshev's inequality outside a fixed neighborhood of τ , strong continuity inside it, and the contraction bound give the stated strong convergence. No spectral diagonalization of the perturbed coefficient generator is presumed. Applying scalar Vitali convergence to every bounded functional and then the locally bounded weak-to-strong holomorphy principle proves analytic dependence of $E_x(\tau)$. Equivalently the sectorial resolvent obtained from (U16) gives the same contour construction. This concerns source analyticity, not a holomorphic continuation of physical time across arbitrary rays.

For real x the full physical heat return is

$$T_x(\tau)F = E_x(\tau)Q_{HF} + \mu_x(F) - \mu_x(E_x(\tau)Q_{HF}). \quad (U25)$$

On the dense smooth domain it solves the actual equation with generator $K - D_v$, preserves its original mean, and agrees with the physical Hilbert semigroup by uniqueness. Its positive physical time is τ/κ . The same formula defines the analytic continuation for complex x . In particular

$$\begin{aligned} & ||T_x(\tau)F - \mu_x(F)||_X \\ & \leq \exp[-d(r)\tau] ||Q_{HF}||_X / (1-\chi). \end{aligned} \quad (U26)$$

Every physical eigenfunction in a finite box is smooth and belongs to Y_0 after subtracting its Haar constant. This last membership follows from Plancherel and Cauchy–Schwarz with a sufficiently high original Casimir power: $\sum_j d_j^2(1+c_j)^{-s} < \infty$ for $s > 3|E_L|/2$. Thus (U24) applied to the exact eigen-equation proves $A = H - E_0 \geq \kappa d(r)$ on the complete centered physical Hilbert space. Compact spectral resolution gives the full form-domain statement. This is a rederivation of the elementary strong-coupling lower bound on this explicit domain, not of every stronger historical source estimate.

D.7 U7. The row-sum estimate that removes the volume factor

For fixed original p put $q_{\text{tau}} = E_x(\text{tau})W_p$ and $h_{\text{tau}} = K^{(-1)}S q_{\text{tau}}$. The exact mean equation gives

$$(K - D_v)h_{\text{tau}} = T_x(\text{tau})W_p - \mu_x(W_p). \quad (\text{U27})$$

For arbitrary complex coefficients z_q with $\max_q |z_q| \leq 1$, set $G_z = \sum_q z_q W_q$. The original edge incidences and trace coefficient 8 give

$$\sup_e \sum_{(q \text{ containing } e)} (1/2) \|z_q W_q\|_X \leq 16.$$

Consequently, with every generator retained,

$$\begin{aligned} & \| \Gamma(h, G_z) \|_X \\ & \leq 3 \cdot 16 \sum_j \sum_e |j_e| \|A_j(h)\|_1 \\ & \leq 32 \|Kh\|_X. \end{aligned} \quad (\text{U28})$$

Use (U21), $\|\mu_x\| \leq 1$, $\|S\| \leq 1/(1-\chi)$, and $\|W_p\|_X = 8$. The connected analytic heat matrix satisfies

$$\begin{aligned} \text{Chat}_{pq}(\text{tau}; x) &= \mu_x[(T_x(\text{tau})W_p - \mu_x W_p)W_q], \\ \sum_q |\text{Chat}_{pq}(\text{tau}; x)| \\ & \leq [256/(1-\chi(r))] \exp[-d(r)\text{tau}] \\ & \leq 512 \exp(-3\text{tau}/2). \end{aligned} \quad (\text{U29})$$

The sum equality to a supremum over phases z_q is exact in every original finite face set. At real x this is precisely $\langle r_p, \exp(-\text{tau} A/\kappa) r_q \rangle$. Its transpose symmetry extends to complex x by analyticity from the real multivariate polydisk. Thus the same bound holds for columns and for the coefficient-space operator norm. No M is introduced.

There is a useful sharper bound that uses the original Haar projection before estimating the returned mean. The physical Fourier block with spin one-half on the four edges of a specified elementary plaquette and zero on all other edges is one-dimensional: at each two-edge vertex the invariant contraction is unique, and following the original indices gives W_p . Consequently, for physical h in Y_0 ,

$$\|P_H \Gamma(h, G_z)\| \leq (1/8) \|Kh\|_X.$$

Indeed the contributing block is $a_p W_p$, has norm $8|a_p|$ and Casimir 3, and its Haar pairing with W_p is a_p . All other product-spin labels integrate to zero. On the nonconstant output, (U18) gives $|\langle \mu - P_H \rangle QF| \leq \chi \|QF\|_X / (1-\chi)$. Retaining both terms in this mean decomposition and using (U28) yields the additional full-row bound

$$\begin{aligned} & \| \text{Chat}(\text{tau}; x) \|_{\text{row}} \\ & \leq [(1+255\chi)/(1-\chi)^2] \exp[-3(1-\chi)\text{tau}]. \end{aligned} \quad (\text{U29a})$$

The original bound (U29) remains available. Take the exact original source radius and its derived constants

$$\begin{aligned} R_1 &= 45/4096, \quad \chi(R_1) = 3/8, \\ C_1 &= 6184/25, \quad d_1 = 15/8. \end{aligned}$$

Then (U29a) supplies

$$\begin{aligned} & ||\text{Chat}-C_0-xi^2 \ C_2-xi^4 \ C_4||_{\text{row}} \\ & \leq (6184/25) \exp(-15\tau/8) \\ & \quad (|xi|/R_1)^6/[1-(|xi|/R_1)^2], \ |xi| < R_1. \end{aligned} \quad (\text{U29b})$$

This second circle has its stated smaller radius and sharper prefactor and decay. A scalar minimum of the two error bounds is permitted; no minimum of noncommuting matrices is introduced. For the k th inverse moment its prefactor is $(6184/25)(8/15)^k$. These are used in the strengthened numerical return M10.

D.8 U8. Uniform analytic remainder and all inverse-energy moments

Now restrict $x_p = \zeta$ homogeneously. The original central link map $U_{-}(n,i) \rightarrow (-1)^{\sum_{j < i} n_j} U_{-}(n,i)$ preserves Haar, K and every gauge constraint and sends all W_p to $-W_p$. Both marked traces change sign, so $\text{Chat}(\tau; \zeta)$ is even. Let $\theta = |xi|/R_0$. Cauchy's formula using (U29), first at radius $r < R_0$ and then letting r increase to R_0 , proves in maximum absolute row-sum norm

$$\begin{aligned} & ||\text{Chat}(\tau; xi) - C_0(\tau) - xi^2 \ C_2(\tau) - xi^4 \ C_4(\tau)||_{\text{row}} \\ & \leq 512 \exp(-3\tau/2) \theta^6/(1-\theta^2), \\ & \quad |xi| < R_0, \ \tau \geq 0. \end{aligned} \quad (\text{U30})$$

The original coefficients C_0, C_2, C_4 are exactly the inherited full heat tables: uniqueness of the local analytic eigen/heat equation at $x=0$ gives the identical recurrence and ground-pole subtraction. Their source and original time coordinates are unchanged.

For the original centered columns R and $A = H - E_0$ put

$$\begin{aligned} G_0 &= R * R, \quad G_k = \kappa^k R * A^{(-k)} R \ (k \geq 1), \\ \Phi &= \kappa A^{(-1)} R, \quad \Phi * \Phi = G_2, \quad \Phi * A \ \Phi = \kappa G_1. \end{aligned}$$

The physical spectral theorem and (U30) justify all integrals, including their complete tails:

$$\begin{aligned} G_k &= \int_0^\infty \tau^{(k-1)} \text{Chat}(\tau) d\tau / (k-1)!, \\ ||G_k - \sum_{j=0}^{k-2} xi^{(2j)} \ G_{k-2j}||_{\text{row}} \\ & \leq 512 (2/3)^k \theta^6 / (1-\theta^2), \ k \geq 1. \end{aligned} \quad (\text{U31})$$

For G_0 , (U30) at $\tau=0$ supplies the same formula with $k=0$. A pole $a/(z+c)^n$ contributes $a(k+n-2)!/[(k-1)!(n-1)!c^{k+n-1}]$. These maps use the original κ in both the heat-time integral and every inverse energy.

D.9 U9. Spatial tails and the actual volume limit

The preceding estimates also specify their spatial return. Set all couplings outside a given finite list to zero. The scalar Hilbert tensor factors on disjoint active link components factorize, as do the positive vacuum and semigroup. Their restrictions to the original invariant observables agree with that scalar calculation. A connected marked correlation therefore has zero multivariate coefficient unless its original marks and source faces lie in one edge-connected component. In particular, if the original face-adjacency distance is $d(p,q)$, a degree- n coefficient connecting distinct marks needs $n \geq d(p,q)-1$. Both marks retain their original links.

At degree n , the same coefficient depends only on the n -neighborhood of the union of marked links, where two links are adjacent when they share an original plaquette. This follows either from the tensor factorization or directly from (U8),(U17),(U25): every source insertion attaches its original connected plaquette union through a differentiated edge, K and its resolvent preserve link support, and all orders add to n . The constant and disconnected products are retained until the connected subtraction.

For two boxes containing the N -neighborhood of the same marks, every homogeneous coefficient through N is identical. Equation(U30)'s coefficient bound gives

$$|\text{Chat}_L - \text{Chat}_{L'}| \leq 1024 \exp(-3\tau/2) \theta^{2(\lfloor N/2 \rfloor + 1)} / (1 - \theta^2). \quad (\text{U32})$$

The identical reasoning gives a row tail outside face-distance D by omitting every coefficient with $n < D-1$; n is even. These are explicit spatial bounds, with the physical link length a unaltered.

For clarity the limiting state and dynamics are also constructed. The coefficient mean (U18) is analytic and bounded by $\|F\|_X$, and its Taylor coefficients depend on finite original neighborhoods. Cauchy tails prove full-sequence convergence $\mu_L(F) \rightarrow \mu(F)$ for every cylinder Fourier polynomial on $|x| < R_0$. Real-coupling positivity and mass one pass from the original vacua. For an arbitrary cylinder function use the actual finite vertex Haar average P_G on its incident vertices; the equality $\mu_L(F) = \mu_L(P_G F)$ extends the limiting physical functional consistently to all cylinder functions. Uniform cylinder approximation then gives its gauge-invariant probability lift on the product of original link groups. Both positivity and this gauge extension are the original finite-volume ones.

Equation(U25) has norm at most $3\|F\|_X$ and localized Taylor coefficients. Hence $T_L(\tau)F$ converges uniformly, first on cylinder Fourier polynomials and then on their continuous uniform completion by the real Markov contraction. The resulting T is positive, unital, strongly continuous, and preserves μ . Its semigroup and symmetry laws pass by approximating the intermediate continuous function uniformly with cylinder polynomials. Equations(U24)–(U26) prove uniqueness of its invariant probability on the gauge-invariant observable algebra: integrate the uniform mixing bound and use density. This claim concerns physical observables; it does not assign a unique gauge-variant probability from those observations alone.

The symmetric Markov semigroup has its self-adjoint generator A_∞ on $L^2(\mu)_\text{phys}$. The finite real bounds pass on cylinder functions by the uniform convergence and then by L^2 density, giving

$$A_\infty|_{(1 \text{ perp})} \geq \kappa \, d(|x|). \quad (\text{U33})$$

On an original cylinder Fourier polynomial F , the functions $(K - D_v, L)F$ converge in X to their infinite-label sum: at each source order they involve finite neighborhoods, and the full drift tails are bounded by the convergent local-source tail times $\|KF\|_X$. Passing to the semigroup integral equation proves F belongs to the limiting generator domain with $A_\infty F = \kappa(K - D_v, \infty)F$. In particular $q_\infty(F, G) = \kappa \mu \Gamma(\text{conjugate}(F), G)$ on this cylinder core. This also fixes the kinetic moment in the limiting original energy pairing.

All local time-ordered correlations pass through the same product/semigroup limits. The row tails above construct bounded infinite plaquette Gram operators on $l_2(P_Z)$, with the same bounds (U29)–(U31). Their coefficientwise local limit and norm-bounded convergence on finitely supported vectors give strong convergence on that l_2 space. This is a fixed- a spatial-volume construction on the displayed coupling domain. It makes no assertion of an ultraviolet field or finite continuum mass.

D.10 U10. Primary sources and scope

G. Lumer and R. S. Phillips, Dissipative operators in a Banach space, *Pacific J. Math.*11(1961),679–698, Theorem3.1, pp686–687: dense-domain dissipativity plus actual range surjectivity gives the contraction semigroup. The original theorem pages were inspected. The coefficient-domain verification is (U22)–(U24).

P. Eymard, L'algebre de Fourier d'un groupe localement compact, *Bull. Soc. Math. France*92(1964),181–236: Fourier-algebra antecedent. The needed compact coefficient product and domain estimates are proved here in (U3)–(U9); no claim is made to have re-audited the complete historical paper.

D. Schuette, Zheng Weihong and C. J. Hamer, The Coupled Cluster Method in Hamiltonian Lattice Field Theory, hep-lat/9603026: exponential-vacuum, gauge-invariant character, and excitation antecedents. The original constants and new heat row-sum return are derived above. The predecessor heat catalogue and the point-of-use source paths are in SOURCE_INTAKE.json. Numerical checks certify their declared finite algebra and endpoints, not the written analytical arguments as a formal proof.

E Growing original plaquette families and their full physical complement

21 September 2026. This is the quantitative return of VOLUME_UNIFORM_HEAT.md, with its source, graph, domain and semigroup proofs. Every finite open box $L \geq 2$, spacing $a > 0$, and coupling $g^2 \geq 16$ is included below. All coefficient axes are original plaquettes. Every Hilbert, energy and quotient pairing is stated explicitly. The static and time-dependent coefficient tables are the inherited original calculations, replayed unchanged; the boundary-safe assembly and rational endpoints below are new calculations. The analytic proof is written for review, not a formal certificate.

E.1 M1. Boundary-safe full-family constants

Write $R0=3/256$, $\theta=\xi/R0$, and $t6=\theta^6/(1-\theta^2)$. The complete marked heat tables have 199 connected multisets containing the anchor face $(0,0,0;0,1)$: one degree-zero self term; one self and twelve adjacent-pair terms at degree two; and one self, twenty-four ordered repeated-pair, 138 path, twelve common-edge, eight corner and two cube terms at degree four. Their 559 marked-row contributions are all retained.

For each contribution, integrate its actual partial fractions as in U31, or take the original time-zero value. Sum absolute values before different original supports can cancel. Every finite box contains a subset of these original contributions at a specified anchor. Thus the following are bounds for every original row, including every boundary row, independent of box size:

Original matrix	degree2 bound	degree4 bound
G0	149/468	2361994073/4691494080
G1	97/468	376882691/938298816
G2	73313/657072	982069718963833/3919180324550400
Kobs	187/468	586668421/1563831360
J3=G0-6G1+9G2	6779/73008	152338674005989/435464480505600

Call these numbers $B_{-}(k,2), B_{-}(k,4)$, and similarly B_{-K}, B_{-J} . The exact transport for every contribution is the original signed coordinate permutation and translation. The independent audit constructs the 199 original multisets without calling that transport producer, verifies its inverse coordinate formula, and reintegrates each original pole directly. The exact row entries of the existing full L2 matrix lie below these bounds; no assertion that the bulk signed row is the largest boundary row is used.

The kinetic matrix is $Kobs=\kappa^{-1}R^*AR=\langle \Gamma(W_p, W_q) \rangle_{\rho}$. Only $p=q$ or faces sharing an original edge can contribute. At $p=q$ the exact function is $4-W_p^2=3-\chi_1(W_p)$. Its trace-coefficient norm is at most 30: the original four-edge spin-one contraction has norm 27, as can be obtained by contracting its three-dimensional indices, or bounded by the original tensor coefficient with one turn. For adjacent faces the direct derivative estimate U4 gives norm at most $3^*(8/2)^2=48$. There are at most twelve neighbors. Therefore, on the full complex source disk,

$$||Kobs(zeta)||_{row} \leq 30+12*48=606. \quad (M1)$$

The spin-one contraction used for 30 has the same four-link index formula as U10 with the SU(2) invariant dual matrix in dimension 3; its nonzero singular values are 3 with total trace norm 27. In the original magnetic-weight basis its fixed dual matrix is $C_{-}(mn)=(-1)^{(j-m)}\delta_{-}(m,n)$,

$m, n = j, \dots, j$. Thus $\pi_j(U^{(-1)}) = C \pi_j(U)^T C^{(-1)}$; both coefficient-side unitary factors are retained. For the trace of four general d -by- d factors with its last two transposed, the coefficient has $A_{(b,c,e; a,b,e,a)} = 1$. Direct multiplication gives $(A^* A)^2 = d^2 A^* A$ and $\text{Tr}(A^* A) = d^4$, so its trace norm is d^3 . Taking $d=3$ proves 27. Its inverse-link signs are represented by the stated dual matrix, not omitted. The accompanying finite check verifies the dimension2 and dimension3 contraction directly. Alternatively the bound $4+8^2=68$ would give 644 and slightly wider kinetic endpoints.

Combining U30–U31, M1, and the table proves

$$\begin{aligned} ||Gk^{-3}(-k)I|| &\leq \xi^2 B_{(k,2)} + \xi^4 B_{(k,4)} \\ &\quad + 512(2/3)^k t_6 = w_k, \quad k=0,1,2, \\ ||Kobs-3I|| &\leq \xi^2 B_{(K,2)} + \xi^4 B_{(K,4)} + 606 t_6 = w_K, \\ 0 &\leq J3 \leq \beta I, \\ \beta &= \xi^2 B_{(J,2)} + \xi^4 B_{(J,4)} + 4608 t_6. \end{aligned} \quad (M2)$$

The whole-matrix bound follows from equality of row and column absolute norm for the original symmetric real matrices, followed by the l2 Schur bound. The remainder $4608 = 512 + 6 \cdot 512 \cdot (2/3) + 9 \cdot 512 \cdot (2/3)^2$ includes all three original moment errors. The matrix $J3$ is positive because it is exactly $(R-3\Phi)^*(R-3\Phi)$, with $\Phi = \kappa A^{(-1)}R$.

E.2 M2. Actual numerical bounds for every original box

For $\xi \leq 1/1024$, equivalently $g^2 = 16$, $\theta \leq 1/12$. Every term in (M2) is increasing in ξ . Evaluation at the endpoint by exact rational arithmetic gives

$$\begin{aligned} w_0 &< 173/10^6, & w_1 &< 116/10^6, \\ w_2 &< 77/10^6, & w_K &< 205/10^6, \\ \beta &< 1555/10^6. \end{aligned} \quad (M3)$$

The sharper fractions, not these rounded values, are retained in generated/uniform_constants.json. Set

$$\begin{aligned} l_0 &= 1 - w_0, & u_0 &= 1 + w_0, \\ l_1 &= 1/3 - w_1, & u_1 &= 1/3 + w_1, \\ l_2 &= 1/9 - w_2, & u_2 &= 1/9 + w_2, \\ l_K &= 3 - w_K, & u_K &= 3 + w_K, \end{aligned} \quad (M4)$$

using the endpoint rational w values. They are positive. The full physical source proof U26 gives $A \geq \kappa d I$ with

$$d = 117/40, \quad (M5)$$

since at $\xi = 1/1024$ its square-root argument is $11/12 > (19/20)^2$. This is a lower bound on the entire centered physical space, proved without selecting an observation family. All physical dimensions and original energies in (M3)–(M5) remain.

E.3 M3. The original observation and its inverse-energy family

Let $R: C^M \rightarrow H_0$ have columns $rp = (Wp - \langle Wp \rangle \rho) \psi$. On its literal coefficient coordinates put

$$\begin{aligned} \Phi &= \kappa A^{(-1)}R, \\ G_0 &= R^*R, \quad G_1 = R^*\Phi, \quad G_2 = \Phi^*\Phi, \quad E = \Phi^*A \quad \Phi = \kappa G_1. \end{aligned}$$

The inequalities above prove R and Φ injective and their ranges closed. The original observation $O = R^*$ has exact left inverse on the primitive family

$$W = G_1^{(-1)}R^*, \quad W\Phi = I. \quad (M6)$$

The state projection $P_R = R G_0^{(-1)}R^*$ retains

$$\|P_R \Phi c\|^2 = c^* G_1 G_0^{(-1)} G_1 c.$$

Using l_1, l_2, u_0, u_2 from the original matrices proves

$$\begin{aligned} \|P_R \Phi c\|^2 &\geq [l_1^2/(u_0 u_2)] \|\Phi c\|^2 \\ &> (624/625) \|\Phi c\|^2. \end{aligned} \quad (M7)$$

This is the inverse-power observation/minimum-section mechanism of the RH workbench's IK9–12, instantiated here by the exact matrix $G_1^{(-1)}R^*$ and the calculated physical Grams. No arithmetic constant or conductor coefficient enters (M7).

The correction between the minimum-state and minimum-energy sections of this observation is $h_R = (I - P_R)\Phi$. Its original energy is

$$\begin{aligned} h_R^* A h_R \\ = \kappa [G_1 G_0^{(-1)} K_{\text{obs}} G_0^{(-1)} G_1 - G_1]. \end{aligned} \quad (M8)$$

Indeed expansion has two cross terms $-\kappa G_1$ and the positive section term; both are retained. Also $q_A(\Phi c, h_R d) = \kappa \langle R c, h_R d \rangle = 0$. Consequently

$$\begin{aligned} 0 &\leq h_R^* A h_R \\ &\leq [u_1 u_K/10^2 - 1] E < (1/1250) E. \end{aligned} \quad (M9)$$

E.4 M4. The complete surrounding physical space

The state projection onto the primitive family is instead

$$\begin{aligned} P &= \Phi G_2^{(-1)}\Phi^*, \quad Q = I - P, \\ B &= QR: C^M \rightarrow QH_0. \end{aligned} \quad (M10)$$

It gives the exact leakage Gram

$$B^*B = G_0 - G_1 G_2^{(-1)}G_1. \quad (M11)$$

For each c , $P R c$ is the state minimum over Φx . Inserting the particular original-coordinate trial $x = 3c$ gives

$$0 \leq B^*B \leq (R - 3\Phi)^*(R - 3\Phi) = J_3 \leq \beta I. \quad (M12)$$

No commutativity of G_0, G_1, G_2 is used in this argument.

Every original form vector decomposes as $\Phi c + h$, h in QH_0 . Its state norm and complete energy are

$$\begin{aligned} ||\text{Phi } c+h||^2 &= c^2 + ||h||^2, \\ q_A(\text{Phi } c+h) &= c^2 E + 2\kappa \text{Re}\langle Bc, h \rangle + q_A(h). \end{aligned} \quad (\text{M13})$$

Since $q_A(h) \geq \kappa d ||h||^2$, the complete mixed term satisfies

$$\begin{aligned} |2\kappa \text{Re}\langle Bc, h \rangle| \\ \leq 2 \sqrt{\beta/(d-1)} \sqrt{c^2 E + q_A(h)}. \end{aligned} \quad (\text{M14})$$

The actual endpoint fractions give $\beta/(d-1) < 1/625$. Thus every vector in the entire physical form domain satisfies the coupled comparison

$$\begin{aligned} (24/25) [c^2 E + q_A(h)] \\ \leq q_A(\text{Phi } c+h) \\ \leq (26/25) [c^2 E + q_A(h)]. \end{aligned} \quad (\text{M15})$$

The off-diagonal term is controlled, not set equal to zero. This is a full-space comparison, not a Ritz lower bound on a selected finite subspace.

E.5 M5. The original complementary inverse and complete Schur loss

A $\text{Phi} = \kappa R$ is bounded as a coefficient map. Hence AP is bounded and PA is its adjoint. Subtract the bounded self-adjoint off-diagonal $QAP + PAQ$ from A . The resulting self-adjoint operator has reducing subspaces PH_0, QH_0 . Its Q restriction is

$$\begin{aligned} D_Q &= QAQ, \text{Dom}(D_Q) = \text{Dom}(A) \cap QH_0, \\ D_Q &\geq \kappa d I. \end{aligned} \quad (\text{M16})$$

The inverse is therefore supplied by the full physical estimate, including every vector of the complementary space. Minimizing M13 over that whole space gives

$$\begin{aligned} h_{\min}(c) &= -\kappa D_Q^{-1} Bc, \\ E_{\text{full}} &= E - \kappa^2 B^* D_Q^{-1} B. \end{aligned} \quad (\text{M17})$$

Testing the residual against all complementary form vectors proves this is the actual minimum, not only a finite-column stationary point. The full loss and restored state metric obey

$$\begin{aligned} 0 &\leq \kappa^2 B^* D_Q^{-1} B \\ &\leq (\kappa \beta/d) I < (1/625) E, \\ (624/625) E &< E_{\text{full}} \leq E, \end{aligned} \quad (\text{M18})$$

$$\begin{aligned} G_{\text{restored}} &= G^2 + \kappa^2 B^* D_Q^{-2} B, \\ G^2 &\leq G_{\text{restored}} < (601/600) G^2. \end{aligned} \quad (\text{M19})$$

The second bound uses $\beta/(d^2-1) < 1/600$. All signs, κ factors, original Grams and complementary feedback are retained. The graph map $c \rightarrow \text{Phi } c + h_{\min}(c)$ has these exact energy and state forms. These estimates address the full complementary coupling left by the predecessor heat construction, with constants independent of both the number of plaquettes and the exterior box.

E.6 M6. Fixed supports, their quotients, and a common physical heat horizon

For original face sets $F \subset G$ put $V_F = \Phi(C^F)$, $U_F = A \setminus V_F = \kappa R(C^F)$. The original complexes are

$$V_F \xrightarrow{A} H_0 \xrightarrow{0} 0.$$

Their support transitions are inclusion in degree 0 and identity in degree 1. The exact transported kernel is

$$V_G/V_F \rightarrow \ker[H_0/U_F \rightarrow H_0/U_G], [h] \rightarrow [Ah]. \quad (M20)$$

Its inverse sends $[Ah]$ to $[h]$. Injectivity of A on H_0 proves the inverse is independent of representatives. Its coefficient-energy minimum, with $J = G \setminus F$, is

$$\begin{aligned} c_F &= -E_{FF}^{(-1)} E_{FJ} \quad c_J, \\ Q_E &= E_{JJ} - E_{JF} E_{FF}^{(-1)} E_{FJ}. \end{aligned} \quad (M21)$$

The removed old-support primitive $\Phi_F c_F$ and both mixed entries remain. The bounds $\kappa \ll I \leq E \leq \kappa u \ll I$ imply exactly the same bounds for every such quotient minimum in its original J coordinates. They follow by minimizing the two complete inequalities on the identical affine fiber.

For $T > 0$ the actual heat columns and forcing defect are

$$\begin{aligned} \Phi_T &= (I - \exp(-TA)) \Phi = \kappa \int_0^T \exp(-tA) R dt, \\ A \Phi_T &= \kappa R - \kappa \exp(-TA) R. \end{aligned} \quad (M22)$$

The omitted primitive is $-\exp(-TA)\Phi$. On the complete centered physical space, $I - \exp(-TA)$ has inverse $\sum_{n \geq 0} \exp(-nTA)$, also on the graph domain, and commutes with A . Thus it gives a cochain isomorphism of every original support diagram. Applying the Split Zero support reconstruction retains its support label and receiving zero together with this actual primitive.

Set $\epsilon = \exp(-\kappa d T)$. Expanding all three heat Gram terms and then applying spectral calculus yields

$$\begin{aligned} (1 - \epsilon)^2 G^2 &\leq \Phi_T^* \Phi_T \leq G^2, \\ (1 - \epsilon)^2 E &\leq \Phi_T^* A \Phi_T \leq E. \end{aligned} \quad (M23)$$

Taking minima over the same fixed fibers carries these inequalities through every finite sequence of original restrictions and quotients, without multiplying a constant for each step. This is precisely the HM19–24 minimum-fiber argument of the RH source, whose coefficient maps remain fixed before the metric comparison. Each rank remains in the determinant bound:

$$\begin{aligned} |L_T - L| &\leq 2B[-\log(1 - \epsilon)] \leq 2B \epsilon / (1 - \epsilon), \\ B &= \sum_j |a_j| \text{rank}_j, \quad L = \sum_j a_j \log \det Q_j. \end{aligned} \quad (M24)$$

Rank-zero terms keep determinant 1. For four signed returns of rank at most 240, the single physical horizon $T = 10/\kappa = 5a/g^2$ gives an error below $4 \cdot 10^{-10}$, in every exterior volume. For ranks that grow with M , a specified tolerance $\epsilon > 0$ is reached at the explicit original physical time

$$T = (\kappa d)^{-1} \log(1 + 2B/\epsilon). \quad (M25)$$

Thus no growing determinant rank has been discarded. The original κ^{rank} factors cancel only between the two determinants in the identical energy frame.

E.7 M7. The actual opposite-face result in every exterior box

Keep $p=(0,0,0;0,1)$, $q=(0,0,1;0,1)$, $\xi=10^{-10}$, $g^2=50000$ and $\kappa=100000/a$. Their inherited complete coefficient $h_4(\tau)$ has zero degrees 0 and 2. All original supports contributing to degree 4 lie inside the $L=2$ box; hence the same h_4 applies in every $L \geq 2$. The new uniform tail U30 gives

$$85910/10^{-7} < C_{pq}(1/\kappa;\xi)/\xi^4 < 85920/10^{-7}. \quad (M26)$$

The previously proved exact coefficient lower bound $h_4(\tau) \geq 5 \exp(-3\tau)/648$, $\tau \geq 1$, is kept. On $1 \leq \tau \leq 3$ the ratio of U30's full remainder to that lower bound is at most

$$\frac{(512 \cdot 648/5) \xi^2 R_0^{-6} \exp(9/2)/[1-(\xi/R_0)^2]}{< 24/1000}. \quad (M27)$$

Thus the actual connected heat correlation is strictly positive throughout $1/\kappa \leq t \leq 3/\kappa$, uniformly over the original exterior volume. The result also passes to the constructed spatial-volume limit U33. These statements use all higher orders, not only the sign of h_4 .

E.8 M8. Infinite plaquette families and the original continuum path

The row-tail and local-limit constructions U9 give bounded operators $G_0, G_1, G_2, Kobs$ on ℓ_2 of all original plaquettes, with exactly the same numerical intervals (M2)–(M5). On finitely supported c , Rc is the original centered local observable in $L_2(\mu)_{phys}$. In this infinite-family statement $Kobs$ denotes the closed-form pairing $\kappa^{-1} q_{\infty}(A_{\infty})(R., R.)$, whose boundedness follows from U33 and the cylinder-generator identity after it. It does not require asserting that $A_{\infty} R$ is a bounded operator on all ℓ_2 coefficients. G_0 proves this map extends boundedly and is bounded below. Define $\Phi = \kappa A_{\infty}^{-1} R$. Then G_1, G_2 are its original mixed and state Grams by the convergent physical heat integrals. They are bounded and bounded below, so $P = \Phi G_2^{-1} \Phi^*$ is again an actual orthogonal projection.

$A_{\infty} \Phi = \kappa R$ is bounded; hence $A_{\infty} P$ and PA_{∞} have the same bounded off-diagonal argument as M16. All maps, complementary minima and inequalities M6–M19 therefore pass to the complete infinite plaquette coefficient space and its full physical complement. This passage uses proved bounded operators, not a formal limit of inverses without a lower bound.

At fixed admissible g, a this controls the spatial-volume limit. For the original simultaneous path $a_n = a_0 2^{-n}$, $g_n^2 = 1/(g_0^{-2} + \beta n \log 2)$, the source disk used in U13 is entered exactly when $c_n = g_0^{-2} + \beta n \log 2 < \sqrt{3}/8$. The narrower numerical metric domain is $c_n \leq 1/16$. For $\beta > 0$ the path eventually leaves each of them. The current result removes the $1/M$ heat radius and the unbounded observation-rank loss on a specified strong-coupling domain; it does not establish a smooth four-dimensional continuum field or finite continuum mass. No universal conclusion is inferred merely from a support kernel or an auxiliary norm.

E.9 M9. What is new in this continuation

The older elementary local source argument U3–U6 has been rederived at its own conservative radius. The new calculation combines its full scalar return with the Poisson identity U21 to bound an entire heat-matrix row, not just one entry. This produces the original box-independent analytic remainder U30–U31. The 199-support absolute assembly then gives original state/response/kinetic metrics for growing families, and M12–M19 controls their full complementary coupling. The spatial and physical maps are explicit throughout. No larger coupling threshold or previously unaudited historical theorem is assigned a new verification status.

E.10 M10. Sharper original-Haar return and broader family domains

The additional bound U29b gives the same complete inverse-moment errors with the smaller circle $R1=45/4096$ and prefactors $(6184/25)(8/15)^k$. For a fixed real x in that disk put $t0=(x/R0)^6/[1-(x/R0)^2]$, $t1=(x/R1)^6/[1-(x/R1)^2]$. Define the actual scalar error bounds

$$\begin{aligned} w_k^{\text{sharp}} &= x^2 B_k(2) + x^4 B_k(4) \\ &\quad + \min(512(2/3)^k t0, (6184/25)(8/15)^k t1), \quad k=0,1,2, \\ \beta^{\text{sharp}} &= x^2 B_J(2) + x^4 B_J(4) \\ &\quad + \min(4608t0, (6184/25)[1+6(8/15)+9(8/15)^2]t1). \end{aligned}$$

The kinetic w_K from M2 is retained. Each scalar expression is increasing in x . Replacing the endpoint l and u values in the already proved M7–M19 inequalities by these evaluated sharper endpoints gives the following complete original-family bounds, all uniform over $L \geq 2$ and $a > 0$:

g^2 at least	observed fraction in M7 exceeds	E_{full}/E exceeds	$G_{\text{restored}}/G2$ is less than	h_R energy / E is less than
10	39/50	18/25	33/25	1/6
12	97/100	243/250	103/100	1/60
16	999/1000	999/1000	1001/1000	1/1900

Here every entry with matrix forms means the displayed Loewner inequality in the same original coefficient coordinates. The full-space gap lower constants used in those three rows are respectively $d=14/5, 72/25, 117/40$ times κ . Each follows directly from U26 by squaring its positive rational comparison. The file generated/sharp_constants.json contains every rational input, interval endpoint and returned fraction; the independent auditor recomputes each from the original marked coefficients.

At $g^2 \geq 16$ the absolute Gram widths obey

$$\begin{aligned} \|G0-I\| &< 124/10^6, \\ \|G1-I/3\| &< 66/10^6, \\ \|G2-I/9\| &< 36/10^6, \\ \|Kobs-3I\| &< 205/10^6, \\ B*B &< (832/10^6) I. \end{aligned}$$

The exact unrounded $\beta^{\text{sharp}}/(d l^{\text{sharp}})$ is smaller than $1/34^2$. Accordingly the full mixed-space comparison M15 sharpens to

$$\begin{aligned} (33/34)[c*E + q_A(h)] &\leq q_A(\Phi c + h) \\ &\leq (35/34)[c*E + q_A(h)]. \end{aligned} \quad (M28)$$

The full complementary minimization loses less than $1/1000$ of E and adds less than $1/1000$ of $G2$ to the restored state norm. No complementary vector or source-kernel direction is removed. The same bounded-operator argument in M8 carries these stronger constants to the infinite original plaquette family. The earlier M2–M27 endpoints remain valid and have their own exact records; this paragraph retains an additional, explicitly proved original-Haar estimate rather than silently replacing their coefficients.